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LAUMED JOURNAL CLUB

LADOKE AKINTOLA UNIVERSITY MEDICAL JOURNAL CLUB

MEDIFICIAL INTELLIGENCE

Revolutionizing Healthcare with Technology

HISTORY OF LAUMED JOURNAL CLUB

The LAUMED JOURNAL CLUB is the journal club of the Ladoke Akintola University Medical Students. It is an integral yet independent arm of LAUMSA made of medical students who have deep-rooted interests in clinical research works, critical thinking and capacity building. On Friday, August the 26th 2016, we had our inaugural meeting which had in attendance the founder: Dr Ayotunde OJO and the co-founders: Dr Sijuade SANUSI, Dr Abimbola OLANREWAJU — all fifth (5th) year medical students then.

The idea that birthed the establishment of the LAUMED JOURNAL CLUB stemmed from the fact that the field of Medicine is growing on a daily basis. It then became important that we, doctors to be, be kept abreast of new innovations and findings in the field of Medicine. It is vital that we keep up with and discuss current literature in a bid to achieve a greater understanding of scientific concepts.

"Drink deep or taste not from the Pierian spring. For shallow draughts intoxicate the brain and drinking largely sobers us again." — Alexander Pope. The saying goes true that a little learning is a dangerous thing. How would one feel to discover after medical school how a little learning here and there can leap us into invalid conclusions? Or what would the feeling be like to discover that we have for the most part succeeded in learning to read and pass examinations? More is definitely better. More in the place of critical appraisal skills, more in critical thinking, more in improved reading habits, more in communication skills, more in group participation and teamwork, more in professional identity, more in the development of research, writing and publishing skills. More, because more is definitely better.

Trousseau said, "The worst man is he who is never an artist, and the worst artist is he who is never a man of science." Medicine itself is a science and an art. The art and the Science of Medicine are complementary. Nowadays, it is sad to see that the medical pendulum is swinging more from the art to the science side. So, the LAUMED JOURNAL CLUB, as we agreed on our inaugural meeting, sought to revive the dying art of Medicine by having a fusion of both science and art so as not to blunt our humanities with too much scientific medicine.

To conclude, Medical practice has over the years experienced and is still experiencing a constant change, the rate of which is accelerating to the point of throwing physicians and students alike off balance. What is accepted as scientific today is discarded by medical practitioners tomorrow in the light of newer evidence. We can learn about new information, new innovations and consequently decide how they modify our practice by promoting interest and capacity building in research, providing a platform for the development of critical thinking skills needed to succeed in this practice. Through a peer-reviewed publication of scholarly journals both in print and electronic forms, the LAUMED JOURNAL CLUB looks forward to gaining academic prestige and notoriety on issues related to the science and art of Medicine.

Special mention and thanks to Dr Muyiwa Adeniran and Dr Joy Oluwaniyi (then Editors-in-Chief of IFEMED and DOKITA respectively). The extensive knowledge and collaboration I enjoyed from them were invaluable at the nascent stage of the LAUMED Journal Club. In the same vein, I remain grateful for the unwavering support and push by the leadership of the LAUMED Journal Club since its inception as led by my humble self, followed by Dr Abimbola Adeagbo, Dr Somtolise Ajufo, Dr Nicholas Aderinto, Dr Kehinde Alare and now the freshly minted Dr Esther Adewuyi, who have all contributed a part in the story that is LAUMED today. Ditto all of our teachers whose shoulders we leaned on for mentorship, guidance, moral and financial support, we are indeed very grateful. We have all played our parts very well in this great making of our club. Thank you all. More is definitely better.

Ayotunde OJO, DR.
Founder & Pioneer Editor-in-Chief, LAUMED Journal Club.

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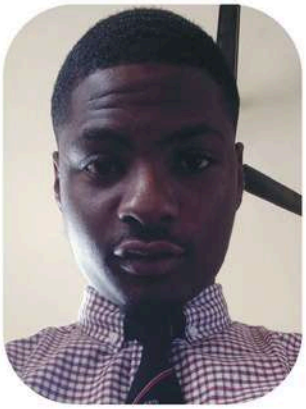
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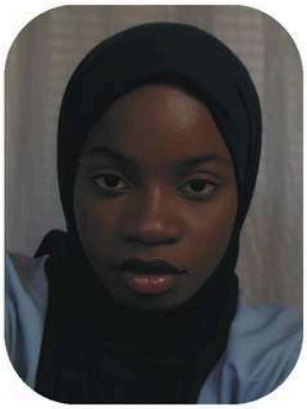
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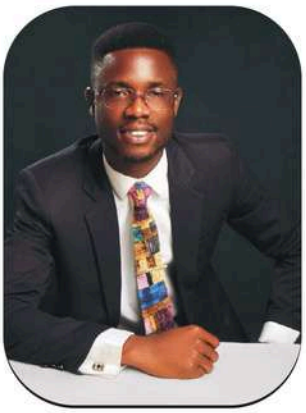
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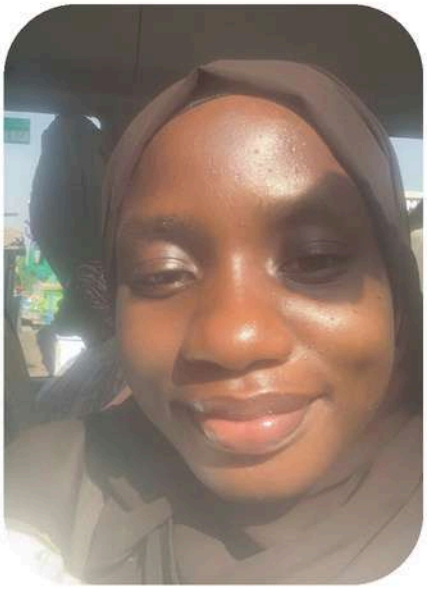
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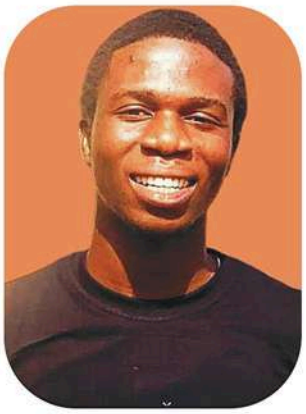
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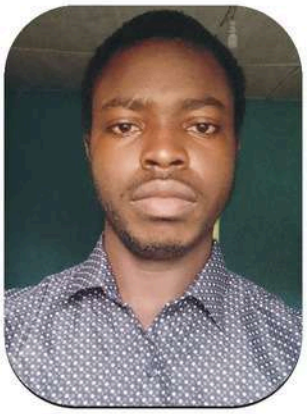
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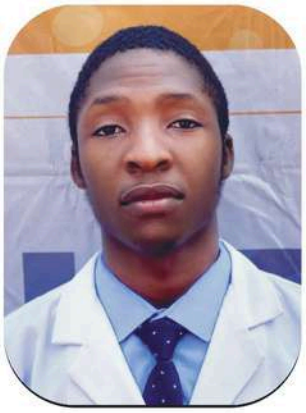
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EDITOR-IN-CHIEF'S NOTE

Medificial Intelligence: Revolutionizing Healthcare with Technology



It is with great delight that I welcome you to the LAUMED 3rd Journal edition, titled, Medificial Intelligence: Revolutionizing Healthcare with Technology.

The fusion of medicine and artificial intelligence (AI) is no longer a distant dream—it is a present reality shaping the future of healthcare. Medificial Intelligence: Revolutionizing Healthcare with Technology delves into this transformation, exploring the impact of AI-driven innovations on medical practice, research, and patient care.

As Editor-in-Chief, I am honored to present this edition of insights, analyses, and breakthroughs that highlight how AI is redefining diagnostics, treatment planning, drug discovery, and healthcare management. This book illuminates the potential and challenges of AI integration in medicine, offering a balanced perspective on its promises and ethical considerations.

From machine learning models that detect diseases with unprecedented accuracy to robotic-assisted surgeries and personalized treatment strategies, each manuscript showcases how AI is revolutionizing healthcare at an accelerating pace. However, alongside these advancements come questions about data privacy, bias, and the evolving role of healthcare professionals in an AI-assisted landscape.
Our goal with this journal book is to spark conversations, inspire innovation, and foster a collaborative approach toward a future where AI enhances, rather than replaces, the human touch in medicine. I invite you to explore these pages with curiosity, critical thinking, and a shared commitment to harnessing technology for the benefit of global health.

To promote a culture of innovation, the LAUMED Journal Club has launched an Innovation Hackathon—an esteemed platform for students to showcase their groundbreaking innovations aimed at advancing healthcare. I am sincerely grateful to Professor Olakulehin, the respected Chief Medical Director of LAUTECH Teaching Hospital, and to the current Patron of LAUMED Journal Club, Professor Michael Olamoyegun for their invaluable support and commitment to this initiative. Also, my gratitude goes to my dedicated board of executives and members of LAUMED Journal for their support towards a successful journal book launch. Thank you.
I welcome you to the future of healthcare.

Esther Adewuyi
Editor-in-Chief, LAUMED Journal Club

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EXPLORING THE ROLE OF ARTIFICIAL INTELLIGENCE IN STROKE TREATMENT AND MANAGEMENT

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ABSTRACT

Stroke is one of the neurological diseases with the most common cause of mortality and disability in the world, significantly impacting individuals' lives, due to its deleterious effect on the brain and nervous system. Due to the time-sensitive nature of stroke, there is a need for prompt and effective treatment methods. Various management strategies have been employed for the management of stroke and have been evaluated in multiple clinical trials. Moreover, the advent of emerging technologies has introduced the use of Artificial intelligence, telemedicine, stem cell therapy, virtual reality, and robotics in the management of stroke and they have shown to be a safe and efficient method of treatment.

This current study provides an outlook on the role of AI in stroke treatment and management through the evaluation of its current state of use in the management of stroke, challenges facing its usage in stroke management, and opportunities for future development in its implementation. We discovered the potential of AI in stroke diagnosis, treatment, and rehabilitation. However, it also presents significant challenges such as AI bias, data privacy, and ethical concerns. Despite these challenges, there are opportunities for growth and development through advancing AI techniques, collaborations between health professionals and AI developers, and identifying potential breakthroughs and research areas to enhance its implementation.

INTRODUCTION

Machine Learning (ML) through pattern recognition algorithms is currently becoming an essential aid for diagnosing, treating, and predicting complications and patient outcomes in several neurological diseases [1]. According to the definition introduced by the World Health Organization in 1970, a stroke is characterized by the rapid onset of clinical signs indicating either focal or global impairment of cerebral function [2]. These symptoms endure for at least 24 hours or result in death, with no identifiable cause other than vascular origin [3]. Stroke ranks as the second leading cause of death globally, resulting in approximately 5.5 million fatalities each year. Each year in the United States, close to 800,000 individuals experience a stroke, while the economic impact of stroke amounts to an estimated $34 billion annually [4]. The morbidity rate is significant, with over half of stroke patients enduring long-term disability [5]. Neuroimaging plays a crucial role in identifying, characterizing, and predicting the outcomes of acute strokes, encompassing both ischemic and hemorrhagic subtypes [1]. Artificial intelligence (AI) technology is rapidly expanding, offering a promising pathway for swift and effective analysis of medical imaging [6] AI technologies have been proposed as suitable tools for implementation in neurology, given the nuanced manifestation of numerous neurological disorders [7]. LVO, the initial CMS-approved AI reimbursement for stroke diagnosis, scrutinizes CT scans to identify early signs of ischemic strokes, promptly notifying the medical staff, thereby reducing the time required for treatment [8]. Numerous AI platforms, both currently operational or in the developmental phase, leverage CT and MRI scans for early stroke detection, treatment, and prognosis assessment [8,9]. As demands rise for healthcare sectors to provide timely, cost-effective, and top-tier medical services, AI emerges as a transformative force with the potential for manifold impacts [10]. AI holds the promise of enhancing diagnostic precision, minimizing errors, and bolstering patient safety by aiding in prescription fulfilment [11].

This current study aims to provide insight into the current state and future direction regarding the application of AI in stroke treatment and management.

CURRENT STATE OF STROKE TREATMENT

Stroke is one of the neurological diseases with the highest mortality rate worldwide. It is a cerebrovascular disease majorly due to the occlusion of an artery which reduces blood supply to the brain causing brain death. Various treatment strategies have been used for the treatment of stroke which can be classified into pharmacological approaches, surgical and neurorehabilitation approaches as well as complementary and alternative medicine.

Pharmacological approaches: the most important pharmacological agent used in treating stroke is the use of antithrombotic agents. Antithrombotic agents such as anticoagulants, antithrombotic drugs, antiplatelet drugs, and thrombolytic drugs are used to prevent arterial thrombosis by promoting intracranial arterial anastomosis as well as limiting and reducing the size of the arterial thrombus in the vessels supplying the affected area of the brain [12].

Anticoagulants such as heparin, dalteparin, and enoxaparin have been useful in the management of venous thrombosis but have limited use in the management of stroke since stroke is caused by arterial thrombosis [13]. However, heparin has been useful in preventing recurrent ischemic stroke although it doesn't have any impact on its clinical outcome [14]. Antiplatelet drugs such as aspirin, aspirin-dipyridamole, clopidogrel, and ticagrelor have a significant impact on the management of stroke and their efficacy has been evaluated in various clinical trials [15-18]. Results from most clinical trials show their efficacy in the treatment of stroke prevention of recurrent stroke, and risk reduction in vascular events and ischemic stroke [16] However, they were associated with various adverse events such as haemorrhage which

can be dangerous in cases of intracranial haemorrhage [17,18]. Antithrombotic medications or thrombolytic therapy have been used in stroke treatment to dissolve arterial thrombus as well as prevent its spread to other regions [19]. One thrombolytic therapy is the use of intravenous alteplase (tPA) which is the widely used and preferred thrombolytic agent in acute ischemic stroke [19]. It is a fibrinolytic agent also known as a tissue plasminogen activator. It has been shown to improve functional outcomes in clinical trials as its implementation within the period of the first few hours of the symptom onset will result in a high treatment outcome for the patient after 90 days [20].

Surgical and rehabilitation approaches: at the prehospital stage i.e. the use of Mobile stroke units (MSU), the patients undergo the preliminary diagnosis to determine the level and severity as well as the nature of the stroke affecting the patients. In the diagnosis process, a non-contrast Computed tomography is used to diagnose the patient. The importance of this neuroimaging modality is its ability to exclude hemorrhagic strokes such as subarachnoid haemorrhage and intracerebral haemorrhage [21]. Another important diagnostic strategy is the use of CT-A which helps in the detection of large vessel occlusion (LVO) and CT-P to determine cerebral blood flow [21,22]. After the diagnosis, the patient either undergoes pharmacological event or surgical treatment depending on the nature of the stroke (ischemic or hemorrhagic). In ischemic stroke surgery, carotid endarterectomy and mechanical thrombectomy are used in their surgical treatment. [23]. Carotid endarterectomy is used to remove fatty plaque in the carotid artery and internal carotid artery [23]. This fatty plague is a result of atherosclerosis which causes narrowing of the arterial wall, reducing blood supply to the brain which results in stroke and transchemic ischemic attack (TIA). Carotid endarterectomy involves a multidisciplinary team that comprises an endovascular surgeon, an endovascular nurse, an anesthesiologist, and a neurosurgeon. Another similar procedure to carotid endarterectomy is the carotid artery stenting which is a minimally invasive procedure that is done endovascularly by imaging procedures to guide a balloon catheter into the carotid artery; the balloon catheter stretches and increases the size of the artery which improves blood flow in the artery [24]. This procedure is said to be more effective than carotid endarterectomy and with lesser risk due to its less invasive procedure. In the management of stroke, craniotomy is used to improve treatment outcomes by opening up the skull to remove a portion of the brain. It is an advanced form of surgery used in the treatment of severe forms of stroke. It is used in relieving intracranial pressure caused by a ruptured blood vessel when treating a hemorrhagic stroke through a decompressive craniectomy while it can be used in removing blood clots when treating an ischemic stroke using the regular craniotomy [25]. They have been widely used in the treatment of stroke due to their safety and efficacy.

AI IN ACUTE STROKE MANAGEMENT

It is noteworthy that Telemedicine has proven to be an effective approach to providing acute stroke care and has facilitated access to large stroke centres [26]. Additionally, tele-neurology visits have been demonstrated to enhance access to general neurologists for outpatient clinical follow-ups in rural locations [27]. It could be revolutionary to support telestroke care with artificial intelligence and machine learning. This includes the ability to diagnose patients through image analysis more quickly and accurately, as well as to forecast their hospital stay and three-month prognosis. Through the analysis of patient charts, Intelligent Electronic Medical Records can do free text searches and offer decision support. Nowadays, speech recognition is efficient and quite practical due to its advancements. Effective alert systems and communication can increase patient flow efficiency and results. Research initiatives and efforts aimed at improving quality can move considerably more quickly and easily when automated data gathering and analysis are used. These synergistic technologies have great potential to improve the clinician-patient experience, lessen physician burnout, and improve patient outcomes at a reduced cost, even despite present obstacles [28]. Artificial intelligence (AI) has revolutionized stroke care through its implementation in various stages of management from diagnosis to rehabilitation (Figure 1) [29-31].

It enhances stroke diagnosis which aids in providing quick, effective, and precise treatment. AI is also a fast-developing field with several uses in acute stroke imaging, encompassing ischemic and hemorrhagic subgroups. Launching timely intervention to lower morbidity and fatality rates requires early detection of acute stroke. Artificial intelligence can assist with several parts of the stroke therapy paradigm, such as prognostication, segmentation, classification, identification of major vascular blockage, grading of the Alberta Stroke Program Early CT Score, and infarct or haemorrhage identification [32]. Particularly, new advances in artificial intelligence, like convolutional neural networks, hold the potential for effectively and precisely completing these imaging-based activities [33]. To assist physicians in the routine and triage management of stroke patients, these models automatically choose pertinent clinical variables, minimize inter-rater variability, and combine neuroimaging with EMR data [34].

Stroke Diagnosis

The clinical management of individuals with stroke requires the use of neuroimaging methods including magnetic resonance imaging (MRI) and computed tomography (CT). Time lags and inter-clinician variations, however, may prevent them from being as useful as possible [34].

In the diagnosis of stroke, AI has been integrated with neuroimaging modalities such as Computed tomography (CT), Magnetic resonance imagery (MRI), CT perfusion imaging, and Positron emission Tomography (PET). The working principle behind its usage is the function of an input dataset, which provides the information and pattern to which the Machine learning algorithm learns which it is going to eventually use in the diagnosis of the patient [33]. AI systems can process data from multiple sources, including imaging investigations, wearable technology, and electronic health records, to give a complete picture of the patient's health. For instance, wearable technology with sensors may track vital signs like blood pressure, heart rate, and oxygen saturation and instantly send the information to medical professionals. These data streams can be analyzed by AI algorithms to find anomalies and forecast possible stroke occurrences, enabling early intervention and ongoing patient monitoring for those who are at risk [35].

Clinical decision support systems (CDSSs) also assist healthcare organizations in analyzing data from electronic health records (EHRs) and making suggestions by providing real-time alerts and reminders to clinicians [36]. A CDSS tool may include an alert for past-due examinations for stroke risk factors. This allows the software to utilize

clinical test outcomes, medical history, and symptoms to determine the likelihood of stroke or to propose therapies for the individual's likelihood of stroke, based on the healthcare provider's decision-making [36,37]. Patient outcomes could be significantly improved by the application of Clinical Decision Support Systems (CDSS) in reducing the risk of stroke. These advantages include greater compliance with evidence-based guidelines for stroke prevention, earlier diagnosis and intervention, and increased accuracy and efficiency in identifying high-risk individuals [38].

Large data sets are pertinent for an efficient function of ML algorithms in the diagnosis of stroke which are limited. This limitation can be attributed to factors such as barriers in data sharing across institutions, "the complexity of building imaging processing pipelines", and "the time and cost of data annotation" [33].

Currently, various machine learning techniques, including logistic regression (LR), linear discriminant analysis (LDA), support vector machines (SVM), Voting classifier (VC), decision trees (DT), random forests (RF), and neural networks have been used in the diagnosis and treatment of stroke [39-41]. These techniques have been very effective in the diagnosis of stroke with Random Forest (RF), decision tree (DT), voting classifier (VC), and logistic regression (LR) being the most successful algorithms used in this respect with an overall accuracy of 92% [40]. These techniques rely on the imputed dataset which helps in identifying neuroimaging-based biomarkers for clinical decision-making during diagnosis.



(Figure 1) [29-31].

Stroke Treatment

Regarding the treatment of stroke, the use of AI can provide personalized recommendations for the best possible treatment strategies for stroke as well as provide prognostic information for its management [42]. Both deep learning and machine learning approaches have been evaluated in the prediction of stroke outcomes in various clinical studies [43-48]. For instance, Fang et. al. (2022) conducted a study on the prediction of functional outcomes after ischemic stroke using deep learning approaches [43]. The study also compared three deep learning approaches (Convolutional Neural network(CNN), Long short-term memory (LSTM), and Residual neural network (Resnet))to the conventional machine learning approaches (Deep Forest, Random Forest, Support Vector Machine, etc.) [43]. Results from the study showed that Both ML and DL are accurate in predicting stroke outcomes. However, Deep learning doesn't outperform Machine learning in the prediction of stroke outcomes which can be attributed to the fact that deep learning approaches are less used in censored and structured medical clinical data [43].

In contrast, results from a clinical trial showed that Deep Learning models outperformed the Machine learning models using traditional radiological image biomarkers in predicting functional outcomes after stroke [44]. The study compared a Deep Learning model, Residual Neural Network (ResNet) architectures to Machine Learning models using traditional radiological image biomarkers.[44] Deep learning models such as multi-CNN, CNN, LSTM, and multilayer perceptron were used to predict poor outcomes in a study by Heo et.al (2020) [45]. When comparing the performance of the DL algorithm with the area under the receiver operating characteristic (AUROC) curve, the multi-CNN algorithm demonstrated the best classification performance (0.805) and was followed by the CNN (0.799) algorithm [45]. The study also showed that DL models outperformed ML models when predicting Future Clinical Outcomes of a Natural Language Processing (NLP)--based ML of radiology free-text reports of brain MRI [45].

Just like in the diagnosis of Stroke, machine learning models such as the support vector machine, random forest, artificial neural network, artificial neural network, extreme gradient boosting, and fully connected deep neural network in addition to the hybrid artificial neural network were used in predicting stroke outcomes in patients. These Models were compared to Assessment scores such as the area under the receiver operating characteristic (AUROC) curve, Houston Intra-arterial Recanalization Therapy (HIAT) score, modified Rankin Scale score (mRS), and the Totaled Health Risks in Vascular Events (THRIVE) score. In Two studies by Lin et.al and Hoe et.al, the machine learning approaches presented high AUC which shows their accuracy as a functional outcome predictor after stroke [45,46]. The former had an AUROC of 0.94 in ischemic and hemorrhagic stroke when using preadmission and inpatient data [46]. On adding data from the follow-up, the prediction ability of the models improved to 0.97AUROC [46]. In the latter, Random Forest was the best predictor of functional outcome with an AUROC of 0.782.[45] Integrated Machine learning models were also compared with traditional risk-scoring models such as the HIAT and THRIVE scores with results showing that they are significantly better than the traditional risk-scoring models when comparing their performance under the AUROC curve [47,48].

AI IN REHABILITATION AND POST-STROKE CARE

When recovering from a stroke, there is a need for neurorehabilitation to improve recovery, reduce disability, and optimize functioning and health outcomes to meet up with the previous state of health. There are various therapies used for neurorehabilitation depending on the area of the body affected by stroke which includes occupational therapy, speech therapy, vision therapy, and language therapy. These therapies require a multidisciplinary team involving a rehabilitation medicine specialist, neurologist, physiotherapist, occupational therapist, pros-

thetist, rehabilitation nurse, psychologist, and vocational counsellor. These health professionals work hand in hand to provide the best possible rehabilitation for the patient and rejuvenate the patient back to his previous state of living.

AI-Driven Personalized Rehabilitation Programs

Personalized medicine is a new branch that provides a targeted therapeutic approach for treating and managing a disease or disorder. It is also a novel medical approach that involves the diagnosis, treatment, and prevention of a disease based on the individual patient's genetic profile [49]. Though personalized medicine does not apply to stroke rehabilitation, using its principles helps in providing a personalized approach to neurorehabilitation patients of stroke using artificial intelligence. AI can help in the personalized neurorehabilitation of stroke by using data from brain scans, language tests, mobility parameters, and videos to predict the recovery potential and tailor the therapy exercises for each patient. AI can also provide patients with feedback, resources, and guidance and monitor their progress over time [50]. AI algorithms using sensor data to make inferences about various patient health outcomes can be classified into supervised learning and unsupervised learning. Supervised learning makes use of labelled datasets from brain scans or language tests to train algorithms to classify or predict data outcomes accurately while unsupervised learning uses algorithms to analyze and cluster unlabeled datasets without any external influences [51]. It focuses mainly on the similarities and differences between the data sets which can help in detecting anomalies in the dataset. AI-powered neurorehabilitation programs for stroke patients often involve the use of robotic coaches and AI. Studies have shown that incorporating robotics into therapy can enhance the number of movement repetitions completed during rehabilitation, which has the potential to improve functional recovery [62,53]. Additionally, it allows for a higher dose of therapy without requiring a therapist. The effectiveness of robotics in rehabilitation has been evaluated in numerous clinical studies [54-59]. In these clinical studies, three studies made use of exoskeleton robots [54-56], two made use of haptic devices [57,58] and one made use of a planar robotic manipulandum [59]. Clinical assessments used in the studies included the Fugl-Meyer Assessment of Upper Extremity (FMA UE), Action Research Arm Test (ARAT), Wolf Motor Function Test (WMFT), modified Ashworth Scale (MAS), and Functional Independence Measure (FIM). Improvements were seen in the Assessment scores in addition to functional and motor recovery in the participants. However, in the study by Hyakutake et.al, there was a disparity in the changes observed in the assessment scores since no significant changes were observed in the FMA scores in some weeks of the study period.[59] Also, adverse effects such as tolerable muscle fatigue were observed in the study by Chen et.al in 2021 [56].

Continuous Monitoring and Feedback through Wearable Devices

Wearable devices such as smartwatches with a built-in accelerometer and Wrist-worn movement sensors have also been applied to monitor rehabilitation exercises in the home setting using datasets to provide feedback and monitor patients' progress over time. This has been evaluated in various studies. For instance, a study by Lee et.al in 2018 on the use of A Wearable Sensor-Based Approach for Upper-Limb Motor Training during neurorehabilitation [60]. The study made use of a wrist-worn sensor system for home-based upper-limb rehabilitation. The system consists of two wearable sensors, a tablet computer to be used by patients at home, and a backend web service for clinician and caregiver data access.[60] Results from the study showed that it was possible to detect goal-directed movements during the performance of Activities of daily living (ADL) with a C-statistic of 87.0% showing the use of supervised learning to measure patient outcomes. In comparison, there were poorly performed movements in selected rehabilitation exercises with an F-score of 84.3% showing the impact of the use of unsupervised learning [60]. Overall, this enabled the generation of appropriate feedback from the participants of the study. An almost similar result was generated in a study by Seo et.al who also evaluated the use of Wearable Sensors to Monitor the Quality of Upper Limb Task Practice for Stroke Survivors at Home [61]. Results from the study showed an accuracy of 93% when classifying the home task practice movement quality while an F1 score of 0.95 for all tasks combined [61]. Another important AI assessment is the use of Fugl-Meyer assessment (FMA) subscores in analyzing the outcome scores of wearable devices in the neurorehabilitation of patients with stroke and it has been evaluated in various studies [62-65]. For instance, a randomized controlled pilot study was conducted to assess the Feasibility and efficacy of wearable devices for upper limb rehabilitation in patients with chronic stroke [62]. The study made use of the Fugl-Meyer assessment, active range of motion, and deviation angle in measuring outcome measures in two groups (device and control groups). All 5 Fugl-Meyer assessment subscores used in the study improved in both the device and control group after the intervention [62]. The other outcome measures also improved substantially after the intervention. Another study by Chae et.al also made use of a wearable device to evaluate its impact on upper limb rehabilitation using the Fugl-Meyer assessment (FMA) as an outcome measure [63]. The study compared FMA with other assessment scores such as the Wolf Motor Function Test (WMFT), power test, Beck Depression Inventory, and range of motion (ROM) [63]. This study differs from the previous one in terms of the technology used. The previous study only used a wearable device with inertial measurement units and wireless sensor networks to monitor motion and give patient feedback. In contrast, this study used a wearable device (smartwatch), smartphone, and custom-developed apps to train a machine learning algorithm to detect home exercises [63]. The device group showed improvement in the mean WFMT scores, ROM of flexion, and internal rotation while there was no significant result in the FMA assessment. This was attributed to the different traits of both functional assessment methods since WMFT is more sensitive than FMA for assessing functional improvement in less affected stroke patients [64]. The potential of a quantitative Fugl-Meyer assessment framework for stroke patients based on a wearable sensor network was expressed in a study by Yu et.al in 2016. In the study, the Assessment can precisely predict FMA scores based on wearable sensor data with a coefficient of determination as high as 0.917 [65]. In the same year, Kim et.al evaluated Functional rehabilitation of the upper limb in hemiplegic stroke patients using depth-sensing cameras in contrast to wearable devices, and the assessment was done using the Fugl-Meyer assessment [66]. Results showed that there was high prediction accuracy and correlation, and, log-transformed jerky scores

were high in combining FMA and Kinect scores for assessment. This also shows its potential for usage in unsupervised home-based rehabilitation [66].

CHALLENGES AND ETHICAL CONSIDERATIONS

Data privacy and security concerns: While incorporating large language models into nursing care planning offers promising advantages, it also brings forth a range of ethical considerations and obstacles. Chief among these is the issue of data privacy and security, given that AI depends on extensive volumes of confidential patient data. Hence, ensuring robust security protocols and maintaining strict data confidentiality are imperative [67]. The utilization of AI algorithms necessitates access to vast amounts of data, which may be susceptible to breaches, unauthorized access, and misuse [68]. Given the extensive scope of dissemination, the concepts of patient confidentiality and privacy may necessitate a complete reimagining [69]. In the aftermath, cybersecurity measures will become increasingly vital for mitigating the risks associated with improper use of datasets, inaccurate or unsuitable disclosures, and deficiencies in deidentification techniques [70]. Transparency regarding both data and AI algorithms emerges as a significant concern, spanning various levels [11].

Ensuring Unbiased AI Algorithms: Algorithmic bias poses a significant ethical challenge in the development and deployment of AI algorithms for stroke care. Biases inherent in training data, algorithmic design, or data preprocessing techniques can lead to disparities in healthcare outcomes, particularly among underserved populations [71]. Addressing bias in AI algorithms requires careful attention to data collection, preprocessing, and model validation processes to ensure the representativeness and fairness of the underlying datasets [72]. Data are not only crucial for the initial training phase but also essential for continual supply to support ongoing training, validation, and enhancement of AI algorithms [11]. To achieve broad implementation, data may require sharing across numerous institutions, potentially spanning international borders. Such data would necessitate anonymization and deidentification, with informed consent procedures addressing the prospect of widespread distribution [11].

Integration with Existing Healthcare Systems: Integrating AI into existing healthcare systems poses numerous challenges related to technical compatibility, interoperability, and organizational readiness. Healthcare institutions often operate within complex IT infrastructures characterized by heterogeneous systems, data silos, and interoperability issues [73]. The seamless integration of AI algorithms into electronic health records (EHRs), clinical workflows, and decision support tools requires collaborative efforts among clinicians, IT professionals, and AI developers to ensure compatibility, usability, and scalability [74].

FUTURE DIRECTIONS AND INNOVATION

Advancements in AI technology for stroke treatment

The widespread of Artificial Intelligence now gives rise to the need for innovations and detailed lookup of stroke solutions in the direction of technological advancement. A systematic literature search was done on a database published from 2012 to 2022 that utilized AI in different areas of stroke care where a total of 623 studies were examined and it was found that the highest AI development area in stroke was in terms of diagnosis (44.1%), and the lowest was in terms of rehabilitation (12%) where a 10-year analysis of trends revealed a significant increase in AI utilization in stroke care [75]. No doubt AI advancement in Innovation creeps has an impact on all cores of disease diagnosis, management, and treatment.

Data Science and Machine learning techniques have shown significant leaps in stroke predictive analysis. A lot of research studies and reviews have been done recently on this which is highly accurate as Saposnik et al. (2013) developed the Stroke Prognostication using the Age and National Institutes of Health Stroke Scale(SPAN) model [76], with an accuracy of 87% in prediction of stroke for '30-day mortality'[76].

Other areas of Predictive Stroke analytics include Early Stroke risk prediction which is very vital to early intervention, hence; the AI algorithm plays an important role in the detection of any signs of risk that could develop into stroke. Brain Scan Interpretation using AI-based Image analysis can also help in identifying the scan based on the type, severity, and location. Moreover, AI can analyze massive amounts of drug data currently on stroke, taking deep features of the drugs such as target, and mechanisms, and paving a new area to identify new drug features which can be of great insights for pharmacologist and aid their development of a better data-driven drug potential. Utilizing large datasets from HER can make Machine learning models identify patterns and correlations not apparent basically with ease

Using tools and IoT (Internet of Things) to get patients' data in real-time, getting continuous data through wearables and sensors to monitor vital signs, and progress, detect adverse cases, and give better monitoring plans [77], which enhances stroke detection and monitoring. Rehabilitation Progress: ML models can dynamically adjust rehabilitation plans ensuring optimal recovery projections [77].

Collaborations Between Healthcare Professionals and AI Developers

Healthcare professionals have the health-based expertise to provide essential clinical insights necessary for channelling developing AI models to stroke [78]. Effective communication between healthcare professionals and AI developers is required to develop solutions that address the needs of patients and are tailored to clinical studies and findings [78]. The need to collaborate between these two ends is very vital as it would enhance the in-depth knowledge in the health case studies i.e. stroke and bring in the expertise of AI. Effective research, communication, collaboration, and sharing of ideas between these people will enhance the development of a solution that comes to the needs of everyone. Collaborative efforts lead to innovative solutions, enhanced clinical plans, and improved patient care.

Potential Breakthroughs and Research Areas

Based on our research, we see potential breakthroughs in the area of interventions for stroke in the areas of Detection and Diagnosis of Stroke using Deep Learning Models and Advanced Neural Network Architecture [79] and AI-based Intelligent Stroke Assistant for Post-Stroke Rehabilitation [80], and other important areas of research to be duly considered include: the ability to integrate reinforcement learning and attention mechanisms to ensure that AI can learn better, improve, and grow in its function in Stroke advancement [77,81]; AI integrates into the EHR system to ensure continuous intelligent monitoring through health data and analytics [82]; extend the study of Artificial Intelligence to improve accuracy, and include additional areas for

advanced imaging, robust data, and detailed fine-tuning for enhanced clinical results [77].

CONCLUSIONS

The use of AI has had a significant impact on the management of stroke, from diagnosis to treatment. It has also proven to be beneficial in neurorehabilitation following stroke management. However, various challenges such as AI bias, data privacy and ethical concerns, security issues, and the integration of this new technology with the healthcare system hinder its use in stroke management. Additionally, there are opportunities for future growth and development through the advancement of AI techniques, collaborations between healthcare professionals and AI developers, and the identification of potential breakthroughs and research areas to further improve its implementation.

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MEASURING BLOOD CHOLESTEROL LEVELS USING WEARABLE DEVICES

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ABSTRACT

The increasing prevalence of diseases associated with abnormal cholesterol levels, like cardiovascular diseases, has emphasized the need for routine monitoring of blood cholesterol levels. Wearable devices, emerging as a promising tool for self-monitoring, offer a convenient and noninvasive alternative to conventional methods of measuring blood cholesterol levels. This review aims to assess the current state of knowledge and recent technological advancements in measuring blood cholesterol levels using wearable devices, exploring their accuracy, reliability, accessibility, affordability, and potential applications. A comprehensive literature search was conducted across various databases, including PubMed and ResearchGate, to identify studies on noninvasive methods of cholesterol monitoring, like wearable devices. The review highlights technological advances in the use of wearable devices to measure blood cholesterol, such as smartwatches. Various technologies, including photoplethysmography, LipidPlus, and Accutrend Plus, have shown promising results in noninvasive methods of detecting changes in cholesterol levels. Wearable devices offer a promising pathway to reducing the risks of cardiovascular diseases. They provide a convenient, accessible and self-monitoring tool for individuals at risk of cardiovascular disease to keep track of their blood cholesterol levels. However, further research is needed to establish the clinical validity, accuracy, and reliability, as well as their cost-effectiveness and potential integration into healthcare systems.

INTRODUCTION

Cholesterol is one of the essential lipids found in the body. It was discovered in 1789 in France and was first isolated from gallstones. It is a 27-carbon molecule structure with a central sterol nucleus of four hydrocarbon rings and a hydroxyl group. Cholesterol is one of the important lipids in the body, serving diverse roles in normal cell functioning. It forms a significant part of the cellular structure in mammals. It is useful in synthesizing steroid hormones, including (neurosteroids and corticosteroids), bile salts, and vitamin D. It is essential in myelin formation and developmental signalling via the hedgehog pathway [1]. Cholesterol metabolism occurs predominantly in the liver and steroidogenic tissues, with other cells serving as modulators of cholesterol content in the membranes by regulating its biosynthesis, uptake, and export from the cells. Cholesterol synthesis in the body occurs through two major pathways: the De novo and exogenous pathways. The de novo pathway occurs in nucleated cells (the liver is the major organ) and involves cholesterol biosynthesis from acetyl-coA molecules. The exogenous pathway involves the uptake of exogenously derived cholesterol from circulation [1].

Low-Density Lipoprotein (LDL) and High-Density Lipoprotein (HDL) cholesterol make up the primary transport forms of cholesterol; other forms, though available, do not contribute significantly to the total body metabolism of cholesterol. LDL cholesterol (bad cholesterol) originates primarily from dietary sources via the liver. It is made up of apoproteins (Apo B100 and Apo E), phospholipids, and cholesterol. LDL is initially packaged in the liver as Very Low-density lipoprotein, its degradation by the enzyme LDL lipase results in the formation of Low-Density Lipoproteins, which are rich in cholesterol.

Low-density lipoprotein is essential for cholesterol utilization from the liver to the peripheral tissues [2]. High-density lipoprotein Cholesterol (also known as good cholesterol) is essential in transporting cholesterol from the peripheral tissues back to the liver and, hence, getting rid of any excess cholesterol from the tissues. Cholesterol, which is amphipathic and lipophilic, is insoluble in the blood and needs apoprotein to enhance its solubility.

THE CLINICAL SIGNIFICANCE OF CHOLESTEROL

Cholesterol is a precursor for producing hormones, especially steroid hormones, including (neurosteroids and corticosteroids), an essential building block for cell membranes and a significant part of the cellular structure in mammals; hence, cholesterol is beneficial to humans to a certain extent. It is also useful in synthesizing bile salts, and vitamin D. LDL transports cholesterol around the body to the parts where it is demanded. However, excess cholesterol in the bloodstream tends to attach to the walls of the arteries and leads to blockage. Deposition of excess cholesterol in the arteries promotes the formation of plaques leading to atherosclerosis. Therefore, LDL is referred to as "bad cholesterol" while "HDL" is referred to as good cholesterol as it gobbles up cholesterol and takes it back to the liver [1,3].

Despite its importance in cellular function, abnormal levels of cholesterol have been observed in various human diseases like heart diseases, stroke, type II diabetes, brain diseases, etc and abnormalities in cholesterol metabolism and levels are associated with significant morbidity and mortality. Low levels of HDL and high levels of LDL are associated with high cardiovascular risks and other disease entities. According to the National Heart Lung and Blood Institute (NHLBI, USA), a higher level of LDL cholesterol increases the chances of heart disease but a higher level of HDL cholesterol reduces the chances of heart disease.

Disease entities associated with abnormal cholesterol levels can either be due to enzyme deficiencies, gene mutation including inborn errors of cholesterol metabolism or defective metabolism such as Antley-Bixler Syndrome, familial hypercholesterolemia, obesity etc. Dysregulated cholesterol metabolism is a risk factor for arteriosclerosis, a major cause of cardiovascular and cerebrovascular diseases worldwide. Other diseases in which cholesterol plays a role in the pathogenesis include cholelithiasis, obesity, myocardial infarction, cerebrovascular ac-

cidents, and fatty liver disease, usually associated with high morbidity and mortality, which are common pathologies that can result from abnormal cholesterol metabolism. Due to the high morbidity and mortality associated with abnormal cholesterol metabolism in the body, patients living with such disease entities are often required to carry out biochemical laboratory tests to measure and quantify their blood cholesterol levels, which can provide a means to determine prognosis from disease processes, the disease progress and to influence the line of management [3,4].

MEASUREMENT OF BLOOD CHOLESTEROL

History of Cholesterol Measurement

The history of cholesterol measurement goes a long way. In 1665, Robert Boyle discovered a fat transport system in animals. In human blood, cholesterol was first discovered in 1833 by Félix-Henri Boudet. The molecular structure was first described by Friedrich Reinitzer in 1888. In 1938, Siegfried Thannhauser and Heinz Magendantz were part of the first to see the relationship between atherosclerosis, xanthomas, and high cholesterol. Using a plasma gel electrophoresis in 1941, Gunnar Blix observed the high amounts of lipids associated with Alpha and beta globulins [5]. Over time, due to technological advancements, there's been an evolution in the methods and techniques through which serum cholesterol levels are measured. In times past, measuring blood cholesterol levels was a sophisticated process that posed a challenge to scientists [6]. However, this challenge has been overcome with increasing technology and scientific advances. Estimating and quantifying cholesterol levels was initially done successfully by plasma gel electrophoresis. However, this was not used for patient investigative purposes but was a part of ongoing research by some scientists. Before that time, cholesterol had just come to the limelight, and various research had been done, forming the basis for what is known today.

Low-density lipoprotein was an initial setback in accurately measuring serum cholesterol levels and was subsequently named Protein X. However, this challenge was overcome by John Goffman and his student Frank Lindgren by using an ultracentrifuge [5].

Why Do We Measure Blood Cholesterol?

In the early 1950s, the association between lipoprotein and cardiovascular disease became clearer. In 1957, Thomas Dawber and William Kannel published a landmark article that shows that higher cholesterol levels increase coronary artery diseases [5]. The association of elevated total cholesterol and cardiovascular disease is one of the leading causes of death worldwide. Therefore, accurate quantification and regular monitoring of cholesterol levels are important for individuals who are at increased risk of these diseases [3].

Cholesterol measurement plays a vital role in the classification of risk factors of individuals and in monitoring the progress of patients being treated. The purpose of treatment is to reduce the LDL cholesterol. Following considerable inaccuracy in cholesterol testing, some recommendations were made on standardization and improvements in cholesterol measurement [5]. Cholesterol are early warning and indicator of various human diseases, cholesterol screening tests are mandatory in adult healthcare and their levels are determined in routine blood tests. In determining and obtaining complete lipoprotein level and profile cholesterol, blood sample is withdrawn after 12 hours of fasting but fasting is not required to determine total cholesterol level [3].

According to the National Cholesterol Education Program (NCEP, USA), optimal LDL and total cholesterol concentrations are <100mg/dL and <200mg/dL respectively. A concentration of HDL <40mg/dL is considered low, while ≥60mg/dL is considered high. The American Heart Association (AHA, USA) recommends all adults age without cardiovascular diseases between 40 and 75 but with LDL-C between 70mg/dL and 189mg/dL to take tests for cholesterol levels every 4 to 6 years [3].

Methods of Biochemical Analysis Of Cholesterol; Advantages And Disadvantage

Over the years, multiple methods have been developed to determine cholesterol levels. In three major categories, they are: 1) Classical chemical methods based on the Abell-Kendall protocol, 2) Fluorometric and colourimetric enzymatic assays, commonly used for assay kits and automated plate readers, and 3) Analytical instrumental approaches such as gas and liquid chromatography or mass spectrometry. Each method has its distinctive advantages. In enzymatic assays, limits of detection are usually low but require costly enzymes. Multi-step procedures are required in classical chemical methods but are relatively simple and inexpensive. The most accurate and sensitive are chromatography and mass spectrometry methods but costly and extensive sample pretreatment. Isotope-dilution mass spectrometry (ID-MS) method and the modified Abell-Kendall protocol are the two approved reference measurement procedures for blood cholesterol quantification by the National Institute of Standard and Technology (NIST, USA) leading to the development of various analytical protocols [7].

Low-density lipoprotein Cholesterol (LDL-C) is the primary risk factor for cardiovascular diseases among the five major lipoprotein classes through atherosclerosis. Hence, it is important to accurately measure the plasma concentration of LDL-C. Various methods are available for measurement of LDL-C which can be classified into analytical methods, calculation methods and automated direct measurement of LDL-C. Analytical methods are considered reference methods and the commonly studied analytical methods are Beta-quantification, vertical spin density gradient ultracentrifugation-based vertical auto-profile, gradient gel electrophoresis, high-performance liquid chromatography, proton nuclear magnetic resonance spectroscopy (1H-NMR), and ion mobility. Unlike analytical methods not used in regular clinical laboratory settings due to non-suitability for automation and, the requirement of experts for operation among many others; automated direct measurement methods are used easily in clinical settings [7].

Technological Advances: The Need for a Better Cholesterol Measurement

Recently, blood cholesterol level has been adequately measured via a fasting lipid profile, which includes LDL and HDL cholesterol and other lipids. These modes of measurement, although accurate, still have some significant drawbacks. Unlike patients living with diabetes mellitus and some other disease entities for whom the advent of portable, self-operational, consumer-friendly machines like the glucometer has made follow-up and monitoring of the disease progression easier, patients with abnormalities of cholesterol metabolism require frequent hospital visits and laboratory testing for adequate follow-up of the disease entity, which can lead to lack of adherence to treatment

secondary to busy schedules, financial constraints and distance of the laboratories to their residence. The advent of wearable devices, in addition to providing a solution to the above socioeconomic factors, also seeks to improve the care of patients as 24-hour, post-prandial, diurnal monitoring is made accessible to the patient.

ADVENT OF WEARABLE DEVICES TO MEASURE CHOLESTEROL LEVELS

Wearable Devices in Healthcare

Over the years, health wearable devices have made rapid strides, evolving from fitness trackers to sophisticated health monitors, revolutionizing healthcare and global health [8]. Wearable devices in healthcare are small electronic devices with requisite sensors to process, analyze, and measure one's health data [9]. These devices monitor the physiological data of people with persistent conditions and facilitate timely medical interventions. They also provide valuable insights into a patient's health, by constantly monitoring health parameters it alerts patients and health care providers of potential health issues shifting the healthcare service from reactive to proactive healthcare [8].

Figure 1 shows wearable devices are used for various medical functions, which include: 1) Safety monitoring for tracking sleep, sweat, heart rate, temperature, and heart attack, 2) Rehabilitation to provide home-based rehabilitation for coronary heart disease patients, other heart disease patients and ageing individuals by recording and monitoring their daily activities, 3) For assessing treatment efficiency and early detection of disorders [10].

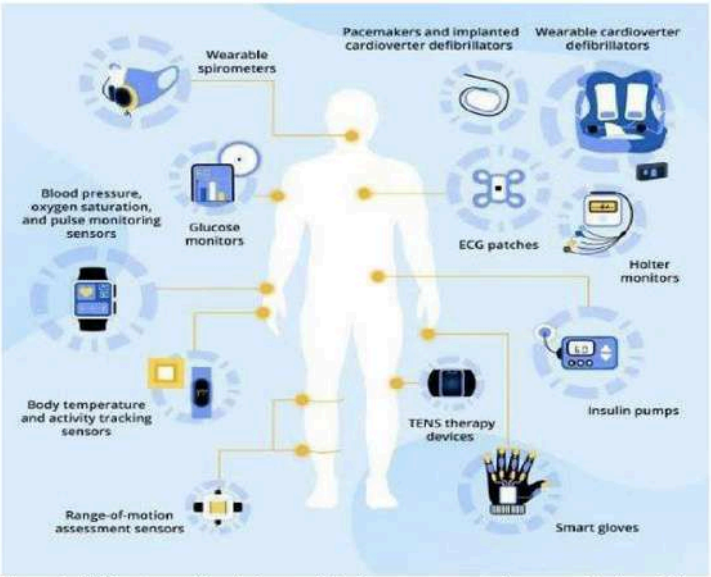


Figure 1: Different wearable devices used in the management of some medical conditions (Adopted from Google image)

Use of Wearable Devices in Measuring Cholesterol Level

High cholesterol level has been a major cause of cardiovascular disease which is a leading cause of premature death globally and knowing your cholesterol level will go a long way to prevent major cardiovascular diseases. Over the years, various methods have been used to measure cholesterol levels which involve various complex methods. There are various cholesterol screening guidelines and tests recommendations which include: The National Cholesterol Education Program guidelines which recommend a fasting lipids panel for adults 20 years and above, every five years; and according to the United States Preventive Task Force, cholesterol tests are recommended every four to six weeks for healthy individuals but individuals aged 35 and above with high risk of cardiovascular diseases should take cholesterol tests more frequently [3,11,12].

The use of non-invasive methods and wearable devices to measure blood cholesterol levels is no longer futuristic as many research and innovations that allow patients to measure their blood cholesterol level individually are currently gaining ground and are better than the invasive and tricky method commonly used. A study by Tzou's team in 2005 showed the strong connection between skin cholesterol levels and serum cholesterol introducing skin cholesterol to assess the risk of cardiovascular diseases. The clinical study by Tzou's team showed that cholesterol accumulation in the skin is associated with the risk of atherosclerotic disease. This method uses a non-invasive, rapid cholesterol detection system utilizing fluorescence spectroscopy to detect the skin cholesterol level without the use of biopsy. It focuses on analyzing how the skin absorbs and scatters light, allowing the scanner to ascertain the skin's default spectral signature, after which the use of a special chemical reagent on the skin which then causes cholesterol present in the skin oil to fluoresce, altering the skin's default special signature. Analyzing the difference between the default and altered signatures is the basis of this method, which allows for determining the skin cholesterol levels corresponding to blood cholesterol levels. Tzou's team concluded that using skin cholesterol as a marker could enhance lipid management in patients with atherosclerosis. This method has been used tested on 121 patients with acute coronary syndrome of whom blood cholesterol level has been measured through traditional methods and proved accurate [13,14].

In another research, a Wireless and soft smart contact lens that enables real-time quantitative recording of cholesterol in tear fluid was used to measure patients' cholesterol levels. This electrochemical cholesterol biosensor on the contact lens platform showed selective, sensitive and reliable sensing performance of tear cholesterol. This research shows the correlation between cholesterol levels in blood and tears, opening a new world to the feasibility of using tear fluids to measure blood cholesterol [11].

Other research and Innovations are still ongoing on the use of wearable devices or non-invasive methods to measure blood cholesterol levels. Various innovations like the use of smartphones to measure cholesterol levels, lipid profiling handheld machines like LipidPlus, Accutrend Plus and many more are gaining ground in the use of non-invasive methods of measuring blood cholesterol [15,16].

Limitations to the Use of Wearable Devices to Measure Blood Cholesterol Levels

Despite the promising performance of the use of wearable devices in measuring blood cholesterol levels, there are limitations to the use and applications. Lack of patient privacy, inaccuracy and non-reliability in wearable devices due to program error or glitches, accessibility and affordability among others are notable limitations to their use [17].

CONCLUSIONS

Wearable devices in measuring blood cholesterol levels are a promising pathway to reducing the risks of cardiovascular diseases. The use of wearable devices can continuously and accurately monitor blood cholesterol levels. Wearable devices have gained ground in various health sectors, particularly in cardiovascular care, like detecting arrhythmias, Blood Pressure measurement and many more, which have proved to help treat and manage cardiovascular diseases. The use of wearable devices to measure cholesterol levels has great promise in cardiovascular and atherosclerosis risk assessment. It creates a convenient, accurate and ideal solution for healthcare providers to monitor cardiovascular-risk patients. However, the use and applications of wearable devices still pose some challenges. Effective Regulatory policies must supervise wearable devices' data to ensure data security. Further studies should be conducted to enable the use of wearable devices that can provide quality, accurate and reliable cardiovascular healthcare. In addition, wearable devices should be affordable and cost-effective.

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THE SIGNIFICANCE OF ROBOTIC SURGERY FOR DEVELOPING COUNTRIES: A FOCUS ON NIGERIA

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ABSTRACT

Robotic surgery represents a significant advancement in Medicine, especially in various surgical specialties. This study focuses on the relevance of robotics in surgical specialties. It involves the use of robots to assist surgeons in performing complex procedures, leading to an exponential rise in the degree of accuracy and precision. While robotic-assisted surgery has gained wide implementation in developed countries, developing countries are still way behind as its' implementation into healthcare systems in LMICs (Low-and-Middle-Income Countries) remains limited due to constraints in terms of financial, infrastructural, and human resources. This review aims to assess the feasibility, challenges, and benefits of implementing robotic surgery in developing countries like Nigeria while also examining the current global state of robotic surgery. We also examine possible economic implications, including cost-effectiveness, scalability, and long-term sustainability. In addition, strategies to stimulate the widespread adoption of robotics in LMICs are also dissected, and they include international collaboration, government policies, public awareness, and NGO intervention. By dwelling on case studies and real-life applications, this study provides insights into how robotic surgery can promote global health equity and reduce morbidity, mortality, and physician burnout. Current barriers to implementation and how to overcome them are also discussed, in addition to prospects and recommendations across all societal levels in the quest towards ensuring global robotic-assisted surgery implementation.

INTRODUCTION

Robotic surgery has emerged as a revolutionary advancement in the field of medicine, characterized by the use of robotic systems to assist surgeons in performing complex procedures with enhanced precision and control [1]. However, the integration of this cutting-edge technology into the healthcare systems of developing countries remains limited, hindered by numerous challenges, including infrastructure deficits, high costs, and a shortage of skilled healthcare professionals [2]. Healthcare systems in many developing regions struggle with inadequate resources and are often overwhelmed by the demand for basic medical services [3]. In these contexts, the introduction of advanced surgical technologies like robotics could significantly improve healthcare delivery by addressing critical shortages in specialist surgeons and improving surgical outcomes through minimally invasive procedures [4].

Exploring the impact of robotic surgery in developing countries is crucial for several reasons. First, it addresses the need to adapt advanced medical technologies to low-resource settings, which could potentially transform surgical care in these regions [5]. Robotic surgery, with its promise of enhanced precision and reduced recovery times, could mitigate some of the pressing healthcare challenges faced by developing countries, such as high rates of postoperative complications and the limited availability of specialized surgical expertise [6]. Furthermore, the successful implementation of robotic surgery in these settings could contribute to global health equity by bridging the gap in surgical care quality between high-income and low-income countries [7]. This exploration is also important for understanding the economic implications of adopting such technologies, particularly in terms of cost-effectiveness and sustainability in resource-constrained environments [8]. As developing countries continue to seek innovative solutions to improve their healthcare systems, evaluating the role of robotic surgery becomes a critical area of focus.

This review aims to assess the feasibility, challenges, and benefits of implementing robotic surgery in Nigeria. By analyzing these factors responsible for the challenges, the review seeks to provide a comprehensive understanding of how robotic surgery can be effectively integrated into the healthcare systems of Nigeria, ultimately improving surgical outcomes and contributing to the broader goal of global health equity.

HISTORICAL DEVELOPMENTS AND APPLICATIONS OF ROBOTIC SURGERY IN MEDICINE

Robotic surgery is a computer-endoscope surgical system where the surgeon sits on a console and controls instruments. This translates the action of the surgeon's hand into a series of smaller and more delicate movements of instruments within the patient's body. The robotic arms have surgically-inserted miniature instruments; the robotic arms can pivot and perform in ways that a human hand cannot, and this provides highly accurate operations even in sensitive and unreachable parts of the body.

The most commonly used system is the da Vinci Surgical System, which consists of three main components: this is a console where the surgeon is seated, a patient side cart that contains four robotic arms that interact with the patient, and lastly is a high-definition three-dimensional vision system [9]. The surgeon's station has operation levers that actuate the robotic arms that contain the surgically necessary instruments and a scope. Some of the features that can be achieved using the system include better dexterity, precision, and vision. Unlike the operation in which the movements of the surgeon's hand are directly translated to the instruments used in this technology, robotic operations translate the movements of the surgeon's hand to robotic arms used in the surgery in a manner that minimizes the effects of the surgeon's tremor, an aspect that makes it possible to perform delicate operations that require higher precision.

The history of redlining in the field of robotic surgery is segmented into the following milestones. The idea of surgery with robots was

conceived in the 1980s when NASA partnered with the Stanford Research Institute to investigate operational surgical systems for exploitation in space travel. The first robotics technology used in surgery was in 1985, the PUMA 560 was used to insert a needle for a brain biopsy using a CT scan [10]. A new advanced system calls for the 1990s, where systems like the ZEUS Robotic Surgical System were utilized for the primary transatlantic telesurgery in 2001. Nevertheless, the major advancement was initiated with the use of the da Vinci Surgical System in 2000, which was more advanced and friendly than all the others. After that, the application of robotic surgery has been expanded, and further technical progress has been made, for instance, improved visualization, intelligent technologies, and miniaturized and multifunctional robots.

Currently, robotic surgery is incorporated in many surgical fields but is most frequently used in urological surgery, gynecological surgery, and cardiothoracic surgery. In urology, robotic systems are typically used for the performance of prostatectomies, and the results are significantly enhanced in the direction of accuracy and rate of healing [10]. Other surgeries performed on the female reproductive system include the use of robots in hysterectomy and myomectomy, as the technique results in less blood loss and shorter hospitalization duration. In cardiothoracic surgery, the robotic systems support those operations that can be made using minimally invasive techniques but are otherwise invasive, for example, mitral valve repair operations [11]. Apart from these, enhanced uses of robotics in colorectal surgery, head and neck surgery, and transplant surgery are coming to light. Robotic surgery's flexibility and accuracy are some of the factors that are still developing its applicability and bringing the quality of minimally invasive surgery with more precision from the machine.

HISTORICAL DEVELOPMENTS AND APPLICATIONS OF ROBOTIC SURGERY IN MEDICINE

Robotic surgery represents a significant advancement in medical technology, offering enhanced precision and minimally invasive options for various surgical procedures. However, the implementation of robotic surgery in Nigeria faces numerous challenges and barriers that hinder its widespread adoption and effectiveness.

Financial Constraint

Insufficient funding for healthcare in Nigeria is the core of these challenges. Despite the Abuja Declaration, which called for increasing government healthcare spending, Nigeria has yet to allocate the recommended 15% of its annual budget to healthcare. The highest allocation recorded was 5.8% in 2012, and the 2020 budget allocated only 4.14% to healthcare. This persistent underfunding makes it difficult for tertiary hospitals to provide essential surgical services or maintain existing infrastructure, and implementing a robotic program seems unattainable.

Also, Out-of-pocket (OOP) expenses, which constitute the primary method of healthcare financing in Nigeria, further hinder access to robotic surgery. Even if a robotic program were implemented, the substantial costs associated with acquiring and maintaining robotic systems—ranging from $0.5 to $2.5 million, with annual maintenance costs between $80,000 and $170,000—along with additional procedure costs of $3,000 to $5,000, would be prohibitive for many patients. The National Health Insurance Scheme (NHIS) aims to mitigate OOP expenses but covers only a limited percentage of the population, leaving many families vulnerable to catastrophic health spending.

For hospitals, particularly those with limited resources, the high costs of robotic surgery systems can be a major barrier. The initial investment and ongoing expenses for maintenance, instruments, and training can be prohibitive. To address these challenges, hospitals might consider financing options such as leasing or partnerships with equipment manufacturers and conduct cost-benefit analyses to demonstrate the long-term benefits and potential savings associated with robotic surgery [12].

Systemic Healthcare Issues

In Nigeria, poor healthcare leadership and management are central to the country's ongoing healthcare issues. Corruption, lack of political will, poor resource management, and ineffective healthcare programs contribute to a system that is inefficient, poorly regulated, and costly, leading to user dissatisfaction and poor health outcomes. These issues hinder the adoption of robotic surgery in Nigeria [13].

Government-related barriers also impede robotics adoption, including neglect of career development in automation and robotics, insufficient industry-academic partnerships, and lack of support for innovative start-ups. Public trust in new technologies relies on transparent, inclusive, and agile governance. Effective regulation is essential to ensure the safe and beneficial use of innovative technologies, protecting both companies and workers [14].

Inadequate Technical expertise

A robotic surgery program needs a specially trained team led by an experienced surgeon and a high patient volume to stay operational. In a country with limited experience in robotic surgery, it is necessary to bring in skilled surgeons and invest heavily in training local staff, which drives up the overall cost of the program [12]. With out-of-pocket payments being the primary method of healthcare financing and about 70% of the population living below the poverty line, patients are likely to prefer traditional open surgery over the more costly robotic option. This results in low patient volume, which restricts hands-on practice, extends the learning curve for robotic surgery and ultimately renders the program unsustainable and cost-ineffective [1].

Inadequate Technical expertise

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Construction sites are often remote and lack reliable power, telecommunications, and internet connectivity. Additionally, construction activities can disrupt power and internet services, posing significant challenges for innovative tools that depend on stable connectivity and power, such as robots and site monitoring systems. For example, sensors and actuators on-site need real-time data processing, highlighting the need for effective solutions to these connectivity issues.

Robotics can be complex due to its integration of information technology, requiring operators to undergo specialized training to effectively use new technologies and avoid obsolescence in the industry [13].

In surgical settings, robotic systems demand substantial electrical power, making a stable and constant power supply crucial. In Nigeria and other developing countries, the power supply is often unreliable, with frequent outages and surges that can damage surgical robots. Hospitals must frequently check that backup generators are operational to mitigate the risks of power failures, which can be costly. The impact of a power outage during a robotic surgical procedure could be severe, emphasizing the need for reliable power solutions

Strategies for Promoting Robotic Surgery

The importance of robotic surgery (RS) in low-middle income countries (LMICs) like Nigeria, where infections are rife and antibiotic resistance is high, includes reducing the risk of surgical site infection and the risk of transmissible infections to surgeons [15]. Also, in LMICs, the patient population in comparison to surgeons is high, and RS will help reduce physician fatigue while also increasing efficacy. These aforementioned reasons, amongst others, are enough to aid the adoption of RS into the health system. A review of RS in Nigeria found no documented use of RS in the country because it has been hampered by low budget allocation to health, epileptic power supply, and poor leadership [16]. To have a successful RS program, the government needs to approve policies that increase health financing – especially technological advancements like RS, to ensure a stable power supply to health facilities and make leaders more accountable.

There are socioeconomic restraints in LMICs that affect RS availability. To overcome this, international collaboration and funding are needed in the form of universal robotic technology licensing to boost product availability. Furthermore, establishing subsidies for equipment and investment in transportation for remote hospitals from international health organizations and donors will help achieve improved access to RS in LMICs.1 The expertise of local surgeons in RS is equally important for proper use. This can be done through initiatives involving collaboration with high-income country (HIC) health institutions, where LMIC physicians can train and have access to research opportunities in RS to promote RS in their native countries by transferring knowledge gained to others. All spheres of life have been affected by technology, but a challenge LMICs have is the long-term sustainability of infrastructures. Painstaking efforts are therefore needed to ensure RS programs successfully implemented are well maintained and integrated and its knowledge dutifully transferred.

FUTURE PROSPECTS AND RECOMMENDATIONS

Emerging Technologies

Integration of AI and Robotics: As artificial intelligence (AI) and machine learning continue to advance, integrating these technologies with robotic surgery could further enhance surgical precision and outcomes. Developing countries should explore partnerships with tech companies to pilot these innovations, leveraging AI to optimize surgical planning and execution [17].

Telemedicine and Remote Surgery: The development of telemedicine and remote surgery capabilities offers an opportunity for expert surgeons in developed countries to assist or guide surgeries in developing regions. Investing in telecommunication infrastructure could facilitate these collaborations [18].

Recommendations for Stakeholders

Governments and Policy Makers: Governments should create policies that incentivize the adoption of robotic surgery, such as tax breaks or subsidies for healthcare facilities that invest in this technology. Additionally, they should prioritize building and upgrading healthcare infrastructure to support advanced technologies.

Healthcare Providers: Local healthcare providers should advocate for the integration of robotic surgery by demonstrating its benefits through pilot programs and case studies. They should also engage in ongoing dialogue with policymakers to ensure that the technology is accessible and sustainable [19].

International Partners and Donors: International organizations and donors play a crucial role in supporting the adoption of robotic surgery through funding, technical assistance, and training programs. They should focus on creating collaborative initiatives that address the unique challenges faced by developing countries.

Potential Impact on Global Health

Enhancement of Healthcare Systems: Over time, the integration of robotic surgery could lead to a substantial improvement in healthcare systems within developing countries. Enhanced surgical precision and reduced complications may contribute to higher overall quality of care and patient outcomes, helping to build more resilient healthcare infrastructures [20].

Reduction in Healthcare Disparities: By incorporating advanced surgical technologies, developing countries could narrow the gap in healthcare quality between them and developed nations. This could lead to more equitable access to cutting-edge medical care, benefiting populations that have historically had limited access to high-quality healthcare.

Economic Impact: The long-term economic benefits of robotic surgery could be significant. Although the initial costs are high, the reduction in complications, shorter recovery times, and decreased need for follow-up procedures can lead to cost savings. These savings can be reinvested into other areas of healthcare, potentially improving overall health services and infrastructure.

Strengthening of Global Health Collaboration: The adoption of robotic surgery in developing countries may foster greater international collaboration and knowledge exchange. This can lead to stronger global health networks, where expertise and resources are shared more equitably, promoting innovation and improvements in healthcare delivery worldwide.

Training and Capacity Building: As developing countries invest in robotic surgery, they will also build local capacity through training and skill development. This will create a pool of skilled professionals who can not only operate advanced technologies but also contribute to the development of the healthcare sector in their own countries and beyond.

CONCLUSIONS

If properly implemented and under regulated supervision, robotic surgery can transform health care in developing countries like Nigeria. This is not unexpected as it focuses on addressing critical gaps in surgical care, improving precision and patient outcomes, and reducing complications. Despite its proven potential, adoption is minimal

in LMICs, majorly due to the high cost of acquisition and maintenance of robots. Other reasons include poverty, unreliable power supply, and lack of adequate public awareness. The challenges are, however, not insurmountable and can be overcome via strategic investment in health-care budget and infrastructure, international collaboration, and government policies. This connotes that successful implementation of robotic surgery in developing nations requires a multifaceted approach and, if achieved a potential solution for improving surgical outcomes and advancing healthcare equity is globally.

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ARTIFICIAL INTELLIGENCE GUIDED EMERGENCY RESPONSE AND DISASTER MANAGEMENT

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ABSTRACT

Emergency response involves a systematic approach to managing the responsibilities of disaster prevention, preparedness, response, and recovery. In 2020, the 389 reported global natural disasters caused over 15 000 deaths, 98 million infected people, and over US$171Bn losses and the COVID 19 pandemic also recently caused 6.5 million deaths, 623 million confirmed cases, and trillions of dollars in losses to date. Artificial Intelligence (AI), with its remarkable capabilities for visual perception, speech recognition, decision-making, and translation between languages is emerging as a potent tool in disaster prevention, early warning, and response efforts, and with increasing incidence of Emergencies, Crises and Disasters, AI is now more indispensable than ever. The use of AI in emergency response and disaster management developed from its initial use in predictive modelling, using algorithms to analyze historical data and identify patterns to forecast future events. AI now employs real time disaster response using tools such as drone surveillance and satellite imagery analysis to assess disaster impact and coordinate response efforts. The development of autonomous systems such has AI robots have been useful in search and rescue missions in hazardous environments. AI has also been used to model and simulate disaster scenarios to help countries in emergency preparedness and response.
The use of AI in disaster management offers a great advantage over human error and limited capacity while maintaining its precision and accuracy. However, the biggest challenge to the use of AI in emergency response and disaster management still lies with its ethical considerations.

INTRODUCTION

Emergency response refers to an immediate and systematic sequence of actions, taken in response to an unexpected, and often dangerous situation that poses an immediate risk to health, life, property, or environment [1,2]. Emergency response is often intercalated with disaster management, which entails processes involved in effectively preparing for and responding to disasters. It involves strategically organizing resources to lessen the harm that disasters cause. It also involves a systematic approach to managing the responsibilities of disaster prevention, preparedness, response, and recovery [3]. Disaster management essentially covers steps to mitigate risk and damage due to accidents and emergencies.

Artificial Intelligence, defined by the man who coined the term himself, emeritus Stanford Professor John McCarthy in 1955, is "the science and engineering of making intelligent machines" [4]. It is an umbrella term for the theory and development of computer systems able to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages. Quantum leaps in the advancement of Artificial Intelligence have seen it gain renowned and rapid integration into our daily lives.

The far-reaching arms of Artificial intelligence (AI) have crept into the field of Emergency response and disaster management to cement visible cracks that have hindered effective emergency response. Over the years budgets and workforces have remained largely stagnant for many emergency-preparedness and response (EP&R) organizations, which include public health, disaster, and other professionals that help the public prepare for and respond to emergencies. Even in developed nations like the United States, estimates project a need for their public health workforce to grow by 80% [5]. However, the workforce has lost more than 45,000 workers in a little over a decade [6]. Many organizations are strained to unprecedented levels trying to match the enormous scale of disasters while still delivering the individualized care each survivor needs [7].

Common causes of disasters and emergencies like natural disasters, which range from hurricanes and earthquakes to wildfires and floods, have wreaked havoc on our planet for centuries. These catastrophic events can lead to immeasurable loss of life and property, and they often leave communities in ruins. While we cannot fully control or predict them, we can use the power of technology and innovation to mitigate the impact of these disasters. Artificial Intelligence (AI), with its remarkable capabilities, is emerging as a potent tool in disaster prevention, early warning, and response efforts [8]. The use of Artificial Intelligence however provides us with its own unique set of challenges and limitations. Issues like the nature of dataset the AI is trained on and their possible bias [8], privacy and ethical concerns, and the relative novelty of the field leave room for justified skepticism.

Emergencies, crises, and disasters (ECDs) have been increasing in frequency, ferocity and scale and extent of devastating impacts, for example, COVID-19 has caused 6.5 million deaths, 623 million confirmed cases, and trillions of dollars in losses to date [9]. In 2020, the 389 reported global natural disasters caused over 15 000 deaths, 98 million infected people, and over US$171Bn losses [10]. In 2022, Hurricane Ian killed at least 126 people and cost over US$67Bn in the USA, the 2022 Pakistan Flood alone deluged 1/3 of the land, killed over 1700 people, and affected 16 million children [9]. Australian National Disaster Assist database recorded 40 severe natural disasters to date in 2022, compared to 47 in 2021 and 46 in 2020. The 2019–2020 bushfire season killed 34 people and destroyed 10 000 buildings [9]. New ECDs emerge every day and become increasingly common, uncertain but devastating. Technological disasters such as cyberattacks, AI weapons, and information disorder, and other emerging events such as geopolitical unrest and financial crises as the global recession further diversify the classic ECD spectrum [9].

AI has proven invaluable to the field of science due its ability to process large volumes of data and then make informed decisions, in double quick time. Our hope is that such attributes can be utilized in the field of emergency response and disaster management. This review aims to critically examine how Artificial intelligence (AI) can be used to improve emergency response and disaster management, its historical evolution, challenges and lessons learnt, the possibility of human-AI collaboration, ethical considerations and future roles of AI in emergency response.

HISTORICAL DEVELOPMENT OF EMERGENCY RESPONSE STRATEGIES

In ancient times, emergency response was largely informal and community-based. Communities relied on local leaders and collective action to address crises such as fires, floods, or invasions [11]. The Roman Empire, for instance, developed one of the earliest organized firefighting services, the Vigiles, around 6 AD. This force was responsible for both firefighting and maintaining public order during emergencies [11]. The Industrial Revolution in the 18th and 19th centuries marked a significant shift in emergency response strategies. Rapid urbanization led to increased risks of fires, accidents, and public health emergencies [12]. In response, cities began to establish formal fire departments and police forces. For example, the London Fire Engine Establishment was formed in 1833, providing a model for modern firefighting organizations [12]. Additionally, the 19th century saw the development of organized ambulance services, with the first recorded civilian ambulance in Cincinnati in 1865 [13].

The World Wars of the 20th century had a profound impact on emergency response strategies [14]. The large-scale destruction and civilian casualties led to the establishment of more formalized civil defense mechanisms [14,15]. During World War II, many countries developed air raid precaution systems, which included early warning systems, shelters, and organized evacuation procedures [14]. These systems laid the groundwork for modern disaster management practices. Following World War II, the Cold War era brought new challenges, particularly the threat of nuclear warfare [15]. Governments around the world, especially in the United States and the Soviet Union, developed extensive civil defense programs [15]. These programs included public education campaigns, construction of fallout shelters, and the establishment of national emergency management agencies [15,16]. In the United States, the Federal Civil Defense Administration (FCDA) was established in 1950 to coordinate national defense and disaster response efforts [16].

In the late 20th and early 21st centuries, the focus of emergency response strategies shifted towards integration and global cooperation [17]. The creation of the United Nations Office for Disaster Risk Reduction (UNDRR) and technological advancements (such as the development of emergency communication systems, Geographic Information Systems (GIS), and early warning systems), in 1999 highlighted the importance of international collaboration in disaster management [17]. The 9/11 attacks in 2001 and the subsequent emphasis on terrorism also influenced the development of emergency response strategies. Governments began to integrate counter-terrorism efforts with traditional disaster response frameworks, leading to the creation of comprehensive national emergency management agencies. For instance, the establishment of the U.S. Department of Homeland Security (DHS) in 2002 was a significant approach in integrating emergency management with national security [18].

INTRODUCTION OF AI TECHNOLOGIES AND THEIR INTEGRATION OVERTIME

The concept of AI can be traced back to the mid-20th century, with foundational work by pioneers like Alan Turing and John McCarthy. Turing's seminal paper in 1950, "Computing Machinery and Intelligence," introduced the idea of machines capable of simulating human intelligence, laying the groundwork for AI research [19]. In 1956, the term "Artificial Intelligence" was coined by John McCarthy during the Dartmouth Conference, where he and other leading researchers outlined the future research agenda for AI [20]. The first wave of AI research, from the 1950s to the 1970s, focused primarily on symbolic AI, which attempted to model human reasoning through rule-based systems. Early AI programs like the Logic Theorist (1956) and ELIZA (1966) demonstrated the potential for machines to perform tasks like theorem proving and natural language conversation, albeit in limited domains [21]. However, this era also highlighted the limitations of early AI, particularly the difficulty of encoding human knowledge and the computational inefficiency of symbolic systems [22].

The limitations of symbolic AI led to a shift towards machine learning (ML) in the 1980s and 1990s. Unlike symbolic AI, which relied on explicitly programmed rules, ML focuses on developing algorithms that can learn patterns from data. This era saw the development of neural networks, inspired by the human brain's structure, which allowed computers to recognize patterns and make predictions from large datasets. Key developments included the back propagation algorithm, which enabled the training of multi-layer neural networks, and the growing availability of computational power, which made ML more practical [23]. The turn of the century marked a significant acceleration in AI development, driven by the explosion of data and advancements in computational power. The availability of vast amounts of data, coupled with powerful GPUs, enabled the training of deep neural networks, leading to breakthroughs in areas like image recognition, speech processing, and natural language understanding.

Deep learning, a subset of ML, became the dominant approach in AI research and applications, with models like convolutional neural networks (CNNs) achieving unprecedented accuracy in tasks like image classification and object detection [24]. During this period, AI technologies began to integrate more deeply into various industries [25]. In healthcare, AI was used to analyze medical images and assist in diagnosis; in finance, AI-driven algorithms transformed trading strategies and risk management; and in consumer technology, AI-powered virtual assistants like Siri and Alexa became commonplace [25]. As AI continues to advance, its integration into society has become increasingly pervasive and complex. The 2020s have seen the rise of generative AI models, such as GPT-3 and its successors, which can generate human-like text and have a wide range of applications, from content creation to customer service [26]. AI is also being integrated into autonomous systems, such as self-driving cars and drones, pushing the boundaries of what machines can achieve independently [27,28].

EMERGENCE OF AI IN DISASTER MANAGEMENT

The initial use of AI in disaster management focused on predictive modeling, where AI algorithms were employed to analyze historical data and identify patterns that could forecast future events [29]. Early AI systems used machine learning models to predict natural disasters such as hurricanes, earthquakes, and floods by analyzing large datasets, including weather patterns, seismic activity, and historical disaster records [29]. For instance, the development of AI-based weather prediction models in the 1990s allowed meteorologists to predict hurricanes and other severe weather events with greater accuracy, providing critical time for emergency preparedness [30]. As AI technologies advanced, their application expanded from prediction to real-time disaster response [30,31]. AI-driven tools such as drone surveillance and satellite imagery analysis have become essential in assessing disaster impact and coordinating response efforts [31]. During the 2010 Haiti earthquake, AI-powered satellite imagery analysis helped to quickly map affected areas, identify collapsed buildings, and prioritize search and rescue operations [31]. Similarly, AI has been used to analyze social media data during disasters, providing real-time insights into the situation on the ground, which aids in the allocation of resources and the identification of people in need [32].

The integration of AI into disaster response has also led to the development of autonomous systems capable of operating in hazardous environments [32]. For instance, AI-powered robots are increasingly used in search and rescue missions, where they can navigate through debris and reach areas that are inaccessible to human responders [33]. AI's role in disaster management extends beyond immediate response to include disaster recovery and resilience building [26,33]. AI technologies are now being used to assess damage after disasters and to plan reconstruction efforts. For example, AI algorithms can analyze satellite imagery to assess the extent of damage to infrastructure, which helps in prioritizing recovery efforts and in the efficient allocation of resources [26]. In the context of resilience building, AI is being utilized to model and simulate disaster scenarios, allowing governments and organizations to better prepare for future events [34]. By simulating various disaster scenarios, AI can help identify vulnerabilities in infrastructure and communities, leading to more effective disaster preparedness plans and the implementation of mitigation strategies [34,35].

HUMAN-AI COLLABORATION

The use of artificial intelligence (AI) in disaster management has offered a great deal of advantage over human error and limited capacity whilst maintaining its precision and accuracy. AI offers a rather fast processing of large quantities of data and tasks requiring such large resources are automated with precise and accurate results as seen in disaster risk applications [36-39]. This automation is hereby very useful in situational awareness to predict disasters and to manage recovery efforts [36,39]. Repetitive tasks can also be automated, making them more cost effective and efficient [36]. Automation has also encouraged a more effective use of human resources in human judgment, supervision and decision-making [36], and will otherwise not be effective in the absence of this personnel [39].

Likewise, hybrid options combining human computing and artificial intelligence are being considered to improve the human capability in disaster response. An example of the joint Qatar Computing Research Institute (QCRI) and the United Nation's Office for the Coordination of Humanitarian Affairs (OCHA)'s Artificial Intelligence for Disaster response (AIDR) which was designed to enhance the processing capability of disaster responders used in sorting large volumes of social media data in real time through human volunteers used in labeling and machines for automatic classification [40]. This would solve the problem of utilizing crowdsourcing in processing this data, thereby saving years of human time [40]. AI can also collect data from microblogging platforms through digital volunteers who collect user tweets on X in the event of a disaster which are classified would be useful in the location of victims and can help to reduce human loss and economic damage [39,40].

ETHICAL CONSIDERATIONS IN AI GUIDED EMERGENCY RESPONSE

The biggest ethical challenge in the use of AI for emergency response lies with the validation [37,38,41]. Large sets of data are needed to balance the risk of bias and variance to enhance the predictive ability of the AI [37,39]. A standard hold-out validation work well when the full data set is large, sufficient and diverse, but when the data sets does not well represent the future hazards, the hold-out testing may be misleading [37]. In many cases, data may be incomplete, outdated, or unavailable, particularly in remote or underserved regions. For instance, the lack of real-time data during a disaster can hinder the effectiveness of AI systems in providing timely and accurate information [35].

Additionally, data collected from various sources (e.g., satellites, sensors, social media) may be inconsistent or prone to errors. Ensuring the accuracy and reliability of this data is crucial for AI models to function effectively, as inaccurate data can lead to flawed predictions and inefficient resource allocation, which can exacerbate the impact of disasters [29]. Likewise, AI systems often rely on personal data from social media, mobile apps, and other sources to provide real-time insights and predictions. This raises questions about data privacy and consent, as well as the potential for misuse of personal information [26].

There are no evidences to support the effectiveness of AI models in rare or surprising events; determining responsibility for errors or unintended consequences can be complex, especially when AI systems are involved in critical decision-making processes. This has however, prompted a need to balance automation with human supervision to maintain efficiency in these situations [34,36].

CHALLENGES FACED WITH THE USE OF AI IN DISASTER MANAGEMENT

Algorithmic Bias and Fairness

AI systems are susceptible to biases that can affect their performance and fairness. Biases in AI algorithms can arise from biased training data, which may reflect historical inequalities or errors. In the context of disaster management, biased algorithms can lead to unequal distribution of resources and assistance, potentially disadvantaging marginalized communities [42]. For example, if AI models are trained on data from predominantly affluent areas, they may not accurately predict or address the needs of less privileged regions. Addressing

these biases requires careful design and validation of AI systems, as well as ongoing monitoring and adjustment to ensure equitable outcomes [30].

Integration with Existing Systems

Integrating AI technologies with existing disaster management systems presents another significant challenge. Many disaster management agencies and organizations rely on established protocols, systems, and technologies that may not be compatible with new AI tools. Integrating AI systems requires overcoming technical and logistical hurdles, such as ensuring interoperability and managing the transition from traditional methods [33]. Moreover, the integration process often involves training personnel to use new AI systems effectively. This can be a time-consuming and resource-intensive process, particularly in organizations that are already stretched thin during disaster events [32].

Resource and Infrastructure Constraints

Implementing AI in disaster management requires significant resources, including financial investment, technical infrastructure, and human expertise. In many regions, particularly in low-income or developing countries, there may be limited access to the necessary technology and infrastructure to deploy AI systems effectively [31]. Additionally, maintaining and updating AI systems can be costly and require specialized knowledge. Ensuring that AI technologies remain operational and effective over time is essential for their successful application in disaster management [42].

CONCLUSIONS

The use of artificial intelligence in emergency response and disaster management is still evolving, however, its role in the management of ECDs have proven to be indispensable due to the immense capability to process large data rather fast and in augmenting human resources in the prevention of loss of life and economic damage. Ethical concern is however the challenge faced in the response to disaster and emergencies.

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ARTIFICIAL INTELLIGENCE IN RADIOGENOMICS FOR CANCER RISK PREDICTION

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ABSTRACT

Cancer stands as a significant contributor to global mortality rates. A considerable number of individuals are diagnosed at an advanced stage, presenting significant challenges in effectively addressing tumour removal within this population. The utilization of radiogenomics in oncology showcases substantial potential for precision medicine. The interdisciplinary field of radiogenomics integrates genomics and radiological imaging to explore the molecular characteristics of tumours. Traditional cancer risk prediction has been the major prediction model that has been in place since the onset for the prediction of most cancers but a lot of limitations have been in place for this method and a lot of risk factors have emerged and are not included in the model. Radiogenomics is advancing cancer treatment by using genomics to provide individualized and personalised care. The predictive capacity of imaging-derived biomarkers in terms of their robustness, clinical implementability, and reliability. The image-based genomics predictive values yielded positive results, and they were able to predict the epidermal growth factor receptor (EGFR). This demonstrated the value of image-based radiogenomics, which is currently in use, and its capacity for cancer prediction correction. The rapid pace of AI technology development, characterized by innovations in machine learning algorithms, deep learning architectures, and computational power, is poised to revolutionize radiogenomics, thus advancements in AI explainability and interpretability will enhance the trustworthiness of AI-driven radiogenomic models, facilitating their integration into clinical practice.

INTRODUCTION

Cancer stands as a significant contributor to global mortality rates. A considerable number of individuals are diagnosed at an advanced stage, presenting significant challenges in effectively addressing tumour removal within this population. Consequently, the early detection of cancer emerges as a crucial determinant in enhancing patient prognosis. The development of cancer risk prediction models assumes paramount importance, as they facilitate the identification of individuals with heightened susceptibility to cancer. These models consider diverse factors such as genetic predisposition, environmental influences, lifestyle behaviours, and medical background. Through the identification of high-risk individuals, targeted screening programs and interventions can be implemented, optimizing treatment efficacy while mitigating the burden of cancer [1].

In recent years, a groundbreaking direction in cancer research has emerged, which delves into the intricate connection between imaging phenotypes and genomics, often termed radio-genomics and imaging genomics. This innovative field integrates data from radiological imaging with genomic information to decipher the molecular intricacies of tumours. Through correlating imaging traits with genomic changes, radiogenomic analysis provides invaluable insights into tumour biology, heterogeneity, and responses to treatment. Moreover, radiomics has emerged as a robust data-driven methodology capable of addressing various clinically pertinent questions, including diagnosis, prediction, prognosis, and treatment assessment [2].

The utilization of radiogenomics in oncology showcases substantial potential for precision medicine. It merges vast datasets of radiomic features extracted from medical digital images with genetic data obtained from high-throughput genomic sequencing, alongside clinical-epidemiological data, to construct sophisticated mathematical models [2].

The interdisciplinary field of "radiogenomics" integrates genomics and radiological imaging to explore the molecular characteristics of tumours. Its primary objective is to establish connections between genetic alterations within cancer cells and the imaging features observed in radiological scans. This fusion of genomic and radiographic data holds significant promise for enhancing cancer diagnosis, prognosis, and treatment strategies. In striving to unravel the molecular underpinnings of diseases like cancer, radiogenomics amalgamates genomic and radiological imaging data. Through this interdisciplinary approach, it aims to uncover correlations between the genetic changes in cancer cells and the imaging abnormalities detected in radiological scans. Radiogenomics presents a potential pathway for advancing cancer diagnosis, prognosis, and treatment planning techniques, effectively bridging the gap between genomics and radiology [3].

TRADITIONAL CANCER RISK PREDICTION METHODS

Traditional cancer risk prediction has been the major prediction model for the prediction of most cancers. For example, traditional models for most Colorectal cancer (CRC) are not genetic specific with most reports showing that the risk for CRC varies depending on the presence or absence of a germline mutation. In most of the traditional risk prediction models considered, the risk factors and common cancer susceptibility genetic mutations were only included. The genetic susceptibility in the model works on the most well-known cancer gene mutations instead of having the ability to derive genotype-specific cancer risks. This is among the limitations of using traditional prediction risk models. Most genetic models cannot adequately account for any complexities of genetic susceptibility for most cancer genes [4,5]. In a study conducted by Thunyarat et al., some limitations of traditional prediction risk models include differences in the distribution of

risk factors across populations which may affect the generalizability of the model to different populations [5]. Some risk factors identified in recent decades have not been added to the most traditional risk prediction models. This limits the usefulness and predictive values of most of the traditional risk prediction models for most cancers.

RADIOGENOMICS: A BRIDGE BETWEEN RADIOLOGY AND GENOMICS

Radiogenomics is defined as the study of the interplay between imaging features and the genetic profiles of tumours. It merges radiomics, which involves extracting and analyzing extensive imaging data, with genomic insights that uncover the molecular foundations of cancer. By combining these fields, radiogenomics helps create predictive models that support clinical decision-making and tailor treatment plans for cancer patients [2,6]. Beyond mere correlation, radiogenomics encompasses the development of mathematical models that integrate imaging and genomic data. This integration enables researchers to delve into the molecular mechanisms driving tumour development, offering new, evidence-based approaches to cancer characterization and treatment. For instance, research in radiogenomics has shown potential in identifying specific genetic mutations linked to distinct imaging features, thus allowing for non-invasive evaluations of tumour biology [7].

The connection between radiological imaging and genomic data is crucial for advancing personalized medicine. Radiological images, such as MRI and CT scans, offer vital insights into tumour structure and function. When these images are analyzed alongside genomic data—revealing the genetic mutations that influence tumour behaviour—clinicians gain a comprehensive view of a patient's cancer profile. This combination is especially beneficial in oncological practices, where traditional diagnostic methods may fall short of capturing the full complexity of tumours. Radiogenomics bridges the gap between imaging data and genetic information, enhancing the interpretation of imaging results by linking them to the genetic landscape. In some instances, certain imaging characteristics may correlate with the expression of oncogenes or tumour suppressor genes, enabling the enhanced identification of molecular cancer subtypes. This can inform treatment choices, such as selecting targeted therapies that are more likely to be effective based on the patient's specific tumour profile [8].

THE POTENTIAL OF AI IN ENHANCING RADIOGENOMICS

Deciphering the interplay of radiological and genetic/molecular features is of utmost importance in oncology and can be achieved through the fusion of genomics and radiomics features in a unified feature space leading to more precise decision support systems [9]. Machine learning, deep learning, neural networks, natural language processing, computer vision, and cognitive computing are all key components of AI. The main goal of AI is creating technology that helps computers to think independently and intelligently, limiting errors, and enhancing accuracy without replacing the functions of pathologists bringing about a collaborative approach [10]. Radiomics has grown over the years and shown much improved potential for developing non-invasive prognostic and diagnostic methods helping with both developing and identifying biomarkers for treatment, and risk prediction for cancers by associating quantitative imaging features developed from various forms of AI for tumor phenotyping and genomic signatures. This is due in large part to the tremendous growth of AI in recent years [11].

A multicenter study that assessed the 5-year recurrence rate and predictive risk of gastric cancer (GC) in patients treated with radical gastrectomy for locally advanced GC developed a radiomic score that was used as a prognostic factor in the majority of the patients; the use of AI, which amplifies the simplicity of the score's design and use and incorporates it into the nomogram, improved classification accuracy and predictive power in the estimation of Recurrence-Free Survival (RFS) and Overall Survival (OS) compared to the TNM staging system alone [12]. In recent years, AI has presented data-driven examination models that have managed the noteworthy signs of progress in information-processing methods in the radiogenomics of cancer [14]. AI use in the role of radiogenomics has been improved and focused on as it has a lot of potential in early diagnosis, risk prediction, scoring the recurrence rate of tumours and improves the mortality rate associated with most cancers.

CURRENT TRENDS AND CHALLENGES IN RADIOGENOMICS

Research on radiogenomics is advancing cancer treatment by using genomics to provide individualized and personalised care [13]. The predictive capacity of imaging-derived biomarkers in terms of their robustness, clinical implementability, and reliability is examined in a study by Gaia et al. [14]. It demonstrated the impact of certain radiogenomics techniques on non-sall cell lung cancer (NSCLC) outcomes. The prediction of actionable mutations was investigated using Positron Emission Tomography (PET) parameters, visual qualitative CT features, CNN-based techniques, and radiomic features acquired from CT and PET imaging. The image-based genomics predictive values yielded positive results, and they were able to predict the epidermal growth factor receptor (EGFR). This demonstrated the value of image-based radiogenomics, which is currently in use, and its capacity for cancer prediction correction. There have also been positive reports on the prediction of other molecular changes [13,14].

Using Deep Learning (DL) in many cancer predictive risk models is currently another important technique. Though not enough study has been done on DL because of its intricacy, its popularity has begun to develop [9,11]. But thanks to recent practice and research, a computer-aided diagnostic (CAD) method for automatically assessing the malignancy of breast tissue samples can be established, as demonstrated by a study by Soliman et al. (9). The method utilizes three sets of attributes associated with nuclei, colour areas, and textures using histological images. This strategy could improve early diagnosis and strengthen Breast Cancer CAD, which would help to reduce the number of preventable fatalities.

Because gene expression and signalling pathways are so intricate, it can be challenging to reconcile the vast amounts of information obtained from whole-genome sequencing with imaging data. Variations in quantitative imaging characteristics can be attributed to patient characteristics or imaging techniques in addition to gene expression [15,16]. Inter-observer variability poses additional challenges, including problems with manual or semi-automated image segmentation,

a lack of standard methods in most facilities, and even intrinsic issues with scanners. Moreover, the radiomics and genomics features acquired are employed by various AI-based techniques to characterize and analyze cancer.

ROLE OF AI IN RADIOGENOMICS

In recent times, AI has introduced data-driven analysis models, leading to significant advancements in information-processing methodologies within cancer radiogenomics. There have been ongoing and progressive efforts aimed at enhancing AI's diagnostic efficacy to enable its adoption in clinical practice [17,18]. The discovery of artificial neural networks (ANN) and their subsequent development presents computational learning models: machine- and deep-learning ideologies. It is mainly accountable for the development of AI in the field of radiogenomics [19]. Machine learning techniques play a pivotal role in analyzing radiomic and genomic data to extract valuable insights into radiogenomics. These techniques encompass a diverse array of algorithms, such as support vector machines (SVM), random forests, and logistic regression, among others. By harnessing the capabilities of these algorithms, radiogenomic studies can discern patterns and correlations between imaging features and genetic alterations, thereby facilitating cancer risk prediction and treatment planning [3].

Successful Applications in Cancer Risk Prediction

Findings have shown the efficacy of machine learning algorithms in cancer risk prediction using radiogenomic data. These algorithms integrate radiomic features extracted from medical images with genomic information to develop predictive models for assessing cancer risk. Successful applications include the identification of high-risk individuals for targeted screening and the tailoring of personalized treatment strategies, ultimately leading to improved patient outcomes [3].

The rapid pace of AI technology development, characterized by innovations in machine learning algorithms, deep learning architectures, and computational power, is poised to revolutionize radiogenomics. With the advent of novel AI techniques, such as reinforcement learning and generative adversarial networks, the capacity to extract intricate patterns from radiomic and genomic data is expected to expand exponentially. Moreover, advancements in AI explainability and interpretability will enhance the trustworthiness of AI-driven radiogenomic models, facilitating their integration into clinical practice [3]. Looking ahead, AI in radiogenomics is anticipated to evolve towards increasingly personalized and precise approaches.

CONCLUSIONS

Radiogenomics will be essential in early diagnosis of cancers and proper evaluation of tumours. The use of AI in the assessment of cancers came into play years ago and many AI techniques have been used for the assessment of cancer risk, progression and its treatment.

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IMPROVING NON-INVASIVE PROCEDURES IN THE DIAGNOSIS OF RENAL PATHOLOGIES USING ARTIFICIAL INTELLIGENCE

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ABSTRACT

Renal pathologies such as AKI, renal tumors and chronic kidney disease, have high morbidity and mortality rate and they pose a significant global concern for nephrology and public health. Artificial intelligence technologies which include machine learning, deep learning, neural networks and so on, are computational technologies that imitate or emulate human-intelligence supporting processes such as adaptation, reasoning, sensory comprehension and interaction. AI has found application in various healthcare domains, where it is used in clinical diagnosis, patient management and therapeutic planning. In nephrology, the technology has been found useful in early detection and prompt diagnosis of several kidney pathologies, optimizing therapy for patients placed on dialysis or those who underwent kidney transplantation, prediction of prognosis and analysis of biopsy images for pathological diagnosis. Limitations to the use of this amazing technology include, unfamiliarity of nephrologists with medical artificial intelligence, data access and quality, data privacy and security, bias, trustworthiness, ethical and legal issues. While AI holds promise as a non-invasive diagnostic technique in nephrology, challenges must be addressed for its full integration into clinical practice and ethical considerations must be put in place when developing AI-powered diagnostic techniques in the future.

INTRODUCTION

Kidney pathologies are described as a heterogeneous group of diseases affecting the normal structure and functions of the kidney [1]. In 2019, approximately 3.1 million deaths occurred due to kidney dysfunction according to a GBD study [2]. Studies have shown that, as a result of their high rates of morbidity and mortality, kidney diseases such as renal tumors, acute kidney injury (AKI), and chronic kidney disease (CKD) are significant global concerns for nephrology and public health [3]. Therefore, it is crucial to identify kidney pathologies early and take steps to stop them from progressing to kidney failure [4].

Although artificial intelligence (AI) is unquestionably becoming more and more relevant, the term itself lacks a generally accepted meaning [5]. AI is commonly referred to as computational technologies that imitate or emulate human intelligence-supporting processes, such as reasoning, deep learning, adaptation, interaction, and sensory comprehension [5,6]. Machine learning is commonly considered a subclass of AI, which concentrates on giving computer systems the capacity to "learn from data," as opposed to copying other elements of the human intellect, like human emotion or movement [7]. The technology can be broadly classified into supervised learning, unsupervised learning, and reinforcement learning depending on various modeling requirements, and its application in the medical field helps to promote medical innovation, reduce medical costs, and also, bring about improved medical quality [8].

Our everyday lives have undergone a shift in paradigm as a result of the rapidly expanding sub-discipline of machine learning, particularly deep learning, and this has generated excitement in the healthcare industry for years due to its potential to support clinical diagnosis, patient management, and therapeutic planning [7]. Other areas where deep learning has been found useful include computer vision, natural language processing, social media, IoT, and autonomous driving [7,8]. Artificial intelligence (AI) has been applied in several healthcare domains, such as mental health, diabetes, cardiology, cancer research, prognosis identification, Alzheimer's disease identification, clinical group differences identification, cardiovascular disease identification, radiographic imaging, Ophthalmology, drug discovery, stroke-related studies and so on [9]. The treatment and clinical diagnosis of kidney pathologies make extensive use of AI which can help with early diagnosis and prompt therapy for patients with acute kidney injury (AKI), from aetiologies such as contrast-induced nephropathy (CIN), postoperative AKI, and burns [10]. AI can also be useful in identifying and predicting the progression of patients with chronic kidney disease (CKD), especially those with IgA nephropathy (IgAN), diabetic kidney disease (DKD), and autosomal dominant polycystic kidney disease (ADPKD) [11]. Furthermore, AI is capable of predicting the risk and optimizing therapy for individuals undergoing hemodialysis (HD), peritoneal dialysis (PD), and kidney transplantation. Lastly, AI can be used to analyze kidney biopsy images for pathological diagnosis [9].

NON-INVASIVE DIAGNOSTIC TECHNIQUES FOR RENAL PATHOLOGY

Molecular imaging is a subfield of medical imaging science that makes use of extremely sensitive equipment and contrast mechanisms to non-invasively identify, locate, and track important molecular processes and as well measure biological and chemical processes in living things [11-13]. Molecular imaging combines imaging technology with particular molecular probes to allow for the non-invasive and quantitative monitoring of physiological or pathological processes [13]. Depending on the imaging method, these probes could be small molecules, peptides, antibodies, nanobodies, aptamers, or nanoparticles that have been functionally modified to function as radiolabeled tracers, fluorescence-emitting agents, or contrast agents [13]. The commonly used molecular imaging techniques for identifying several diseases include;

computerized tomography (CT), magnetic resonance imaging (MRI), diagnostic ultrasonography (US), bone scan, Positron emission tomography (PET), and plain film x-ray [14,15].

However, the above techniques have some important drawbacks, which include but are not limited to; low resolution, the possibility of radiation injury, the time-consuming and error-prone nature of the interpretation of the images particularly in some complex cases such as kidney cancer or CKD, and the possible danger of enhanced nephrotoxicity due to the use of contrast agent [16,17]. In addition, the methods are not suitable for the early diagnosis of some kidney pathologies such as AKI [16].

Another non-invasive diagnostic technique employed in kidney pathology is urinalysis [18-20]. Non-invasive urinalysis involves the collection of urine by spontaneous voiding and testing the urine for the presence of several biomarkers that could indicate the presence of different renal pathologies [19]. Urinary biomarkers, such as liver-type fatty acid-binding protein (L-FABP), kidney injury molecule-1 (KIM-1), plasma neutrophil gelatinase-correlated lipocalin (NGAL), and interleukin-18 (IL-18), have been demonstrated in numerous studies to be effective in the detection of renal pathologies, such as AKI before a change in serum creatinine [20]. The prognosis and diagnosis of AKI patients may both be enhanced by these biomarkers [20].

SPECIFIC AI TECHNIQUES FOR RENAL PATHOLOGIES

Artificial intelligence (AI) has been incorporated into healthcare in recent years, opening up new avenues for improvement in patient care, diagnosis, treatment, and operational efficiency in healthcare systems. Artificial intelligence (AI) technologies, such as computer vision, natural language processing, machine learning, and deep learning, have revolutionized many facets of healthcare delivery [21]. Young Su Joo et al.(2023)[22], created the "Reti-CKD" score, a non-invasive CKD risk stratification tool, with the theory that small changes in the retinal vasculature could reveal information about a person's likelihood of developing CKD in the future. This is accomplished by combining clinical variables and deep learning algorithms that were trained on 158,216 retinal images. Reti-CKD score outperforms existing eGFR-based approaches in terms of predictive performance and successfully stratifies CKD development risk, as demonstrated by internal and external validation in the UK Biobank and Korean Diabetes Cohorts. In an attempt to assist clinicians in intervening during what may be a critical period for averting permanent kidney impairment, the influence of AI on the early diagnosis and prediction of acute kidney injury (AKI) has been specifically studied [23]. Furthermore, Adhikari et al. (2022) [23], presented "Intraoperative Data Embedded Analytics" (IDEA), a program that uses additional data gathered during surgery and a physiological time series to revise the preoperative risk. In addition, High sensitivity and specificity postoperative AKI predictions were improved by the dynamic inclusion of intraoperative data.

Geddes CC et al.(1998) [24], devised and executed the development of ANN to show the existence of a function that forms a relationship between clinical and biological parameters, such as sex, blood pressure, proteinuria, serum creatinine level, age, and anti-hypertensive therapy at the time of the diagnosis of IgAN and the possibility of developing progressive IgAN. When the results were compared to experienced nephrologists, the ANN performed better at anticipating the start of a progressive IgAN. By utilizing deep learning, multiple machine learning approaches, and the Bayesian classifier—a learning algorithm that relies on the statistical correlation between radiomic and computed tomography (CT) texture analysis—it is now possible to accurately identify the imaging features of kidney mass, to diagnose and also differentiate benign kidney masses from malignant ones without the need for invasive procedures [3,17]. For example, To distinguish renal cell carcinoma (RCC) from benign solid kidney masses (renal oncocytomas and fat-poor renal angiomyolipoma), Zabihollahy et al.(2020) [25], recently used a deep learning model that was developed on contrast-enhanced CT (CECT) images from 155 patients. The model was then validated on CECT images from 160 patients. With the help of the semi-automated approach, the suggested CNN algorithm achieved 83.75% accuracy, 89.05% precision, and 91.73% recall rate. Also, it has been demonstrated that one of the best ways to correctly identify a patient suffering from AKI is via the use of AI techniques [3]. Research has shown that early identification of AKI has many advantages associated with it, hence, several machine learning and deep learning algorithms that are capable of predicting the risk of AKI and sending automated alerts to patients within 48h and 30 days of manifestation, respectively have been developed [5,26]. Shinji Kitamura et. al.(2020) created a deep learning model that could help in the diagnosis of diabetic nephropathy(DN) by utilizing renal Pathological Immunofluorescent Images [10].

CHALLENGES AND ETHICAL CONSIDERATION

Artificial intelligence is playing an increasingly important role in many fields of medicine, assisting physicians in most steps of patient management [27]. In nephrology, artificial intelligence can already be used to improve clinical care, hemodialysis prescriptions, and follow-up of transplant recipients. However, many nephrologists are still unfamiliar with the basic principles of medical artificial intelligence.

AI can improve clinical care and research, but its use requires a solid understanding of its fundamentals, the promises and perils of algorithmic fairness, the barriers and solutions to its clinical implementation, and the pathways to developing an AI-competent workforce. The potential of AI in the field of nephrology is vast, particularly in the areas of diagnosis, treatment, and prediction. One of the most significant advantages of AI is the ability to improve diagnostic accuracy. While AI has tremendous potential, there are also significant challenges to its implementation, including data access and quality, data privacy and security, bias, trustworthiness, computing power, AI integration, and legal issues.

To adopt a new technology, clinicians and scientists need to be familiar with its fundamentals, fluent in its vocabulary and nomenclature, and inspired by its potential to improve outcomes for patients with or at risk of kidney disease. Due to the integration of AI into the field of medicine, nephrologists will soon engage with this technology daily, making it essential for the nephrology community to be well-informed and educated about its implications. Given nephrologists' medical duties and legal responsibilities for the care they provide, they may be understandably reluctant to take action on a patient based on unexplained decisions made by black-box algorithms. The success of

any AI-based study will therefore require strong interdisciplinary collaboration between medical experts and computer scientists, always avoiding the black-box feeling that refers to the opaqueness that is often perplexing. Black-box is directly related to the lack of interpretability of the AI system. The potential for bias in AI algorithms is a further risk that should be considered. If an AI algorithm is trained on biased data, it may perpetuate or even amplify existing biases in the healthcare system. For example, if an AI algorithm is trained on data from predominantly white patients with focal glomerulosclerosis, it may not accurately identify or classify the disease in patients from other racial or ethnic groups. It is human nature to trust only things that are easy to understand, and doctors are a perfect paradigm of this. One of the critical challenges in implementing AI is the unknown nature of how learning models and a set of inputs can predict the output and provide guidance in a medical intervention.

Area	Current Methods	AI Based Improvement	Potential Benefits	Challenges
Imaging Techniques	Ultrasound Magnetic Resonance Imaging Computed Tomography	AI based image analysis and interpretation	Improved accuracies and faster diagnosis	Data quality with existing system
Detection of Anomalies	Manual Examination	Automated detection algorithm	Early and accurate detection	False positive/negative algorithm
Predictive Modeling	Statistical analysis	Machine learning model	Predicts disease progression Reduce invasive tests	Need for large high quality dataset
Patient Monitoring	Regular follow up and check up	AI enabled wearable devices and application for adequate monitoring	Continuous monitoring and early intervention	Device reliability Patient adherence
Cost	High cost due to manual intervention	Reduced cost through automation	Lower health care cost	Initial investment in AI infrastructure
Time Efficiency	Time-consuming manual processes	Real-time analysis and decision-making	Faster patient throughput	Integration with clinical workflows

Table 1: Summary of the benefits and challenges involved in implementing the use of AI in non-invasive procedures in the diagnosis of kidney pathologies.

Another risk to consider is the potential for errors in AI algorithms. While AI algorithms can process large amounts of data quickly and accurately, they are not perfect and can make mistakes. It is important to have systems in place to identify and correct errors in AI algorithms to ensure the safety and effectiveness of the technology. As mentioned above, for the time being, AI tools will help nephrologists make decisions, but will not replace expert professionals. AI algorithms may also violate laws or regulations, exposing the organization to legal challenges.

ETHICAL IMPLICATION OF AI IN THE DIAGNOSIS OF KIDNEY PATHOLOGIES

Societal issues may take many forms, but, more often than not, they entail the design of models from a perspective, that is, models that incorporate human-centered [28]. This is certainly only a partially technical matter. Privacy-preserving ML-based data analysis must deal with the potentially contradictory problem of keeping personal information private while aiming to model it, often to make inferences that will affect a given population. Data anonymity refers to the impossibility of linking personal data with information about the individual that is not meant to be revealed. These are key problems and concerns in the medical and healthcare domains, mainly in the interaction between the public and private sectors.

Ethics and Fairness

Biological intelligence is multi-faceted and responds to the environmental pressures of human societies. Ethics are one of those facets for which AI is still fairly unprepared. Interestingly, this topic has become central to AI discussion in recent years. Needless to say, ethics are also a core concern in medicine and healthcare. Such convergence of interests makes it important to create a clear roadmap for the ethical use of AI and ML in medicine. The application of ML and AI in areas of social relevance must also aspire to be fair. How do we imbue ML algorithms, which are fairness agnostic, with fairness requirements? How do we avoid gender or ethnicity, for instance, unfairly influencing the outcome of a learning algorithm? In the medical domain and in healthcare in particular, where sensible information about the individual may be readily available, how do we ensure that AI- and ML-based decision support tools are not affected by such bias?

Balancing technological advancements with patient's welfare

In the realm of healthcare, AI emerges as a transformative force, reshaping established practices and offering unprecedented advancements. This comprehensive analysis delves into the multifaceted ways AI is revolutionizing healthcare, focusing on its transformative capabilities, inherent challenges, and the crucial ethical complexities entwined in its application. The challenge lies in balancing transparency and accountability amid the intricate algorithms, particularly concerning the interpretability of AI-generated insights. However, balancing these technological advancements with patients' welfare requires careful consideration and ethical frameworks. Concerns arise regard-

ing patient privacy, data security, and algorithmic biases. Safeguarding patients' welfare entails implementing robust data protection measures, ensuring transparency in AI algorithms, and addressing biases in training data to prevent disparities in healthcare delivery.

Ethical considerations also come into play when integrating AI into kidney pathology practice. Healthcare professionals must uphold principles of beneficence, non-maleficence, autonomy, and justice when utilizing AI technologies. Ensuring patient autonomy through informed consent and shared decision-making processes is essential in maintaining trust and respecting patients' rights.

Moreover, continuous monitoring and evaluation of AI systems are imperative to mitigate risks and ensure patient safety. Regular audits, clinical validation studies, and interdisciplinary collaborations are essential to assess the reliability, accuracy, and clinical utility of AI applications in renal pathology.

FUTURE DIRECTIONS IN THE DIAGNOSIS OF RENAL PATHOLOGIES

AI applications in neuropathology have recently become available through several advancing technologies, including (i) the widespread introduction of glass slide scanners, (ii) data servers in pathology departments worldwide, and (iii) greatly improved computer hardware to enable AI training. The future advancements of AI in renal diagnosis hold immense promise, driven by ongoing research and technological innovation. Continued advancements in deep learning algorithms and neural networks will enable AI systems to analyze renal imaging and pathology data with unprecedented accuracy. These systems will be capable of detecting subtle abnormalities and predicting disease progression with high precision [29].

Future AI systems will integrate diverse data sources, including imaging studies, genetic data, biomarkers, and electronic health records, to provide a comprehensive assessment of kidney health. This holistic approach will facilitate earlier detection and more personalized treatment strategies. AI-powered monitoring systems will enable continuous assessment of renal function and response to treatment in real time. These systems will provide clinicians with timely feedback, allowing for early intervention and optimization of patient care. AI-driven telemedicine platforms will facilitate remote consultations and monitoring of patients with kidney conditions. These platforms will leverage AI algorithms to analyze data collected from wearable devices and home monitoring systems, enabling proactive management of chronic kidney disease.

Anticipated Impact on Clinical Practice

Transitioning from glass to digital has opened the field to the development and application of image analysis technology, commonly deep learning methods, to assist pathologists with tissue examination [30]. Nephropathology is poised to leverage this technology to improve precision, accuracy, and efficiency in clinical practice. Recent findings through a multidisciplinary approach, neuropathologists, and computer scientists have made significant recent advances in developing AI technology to identify histological structures within whole slide images (segmentation), quantification of histologic structures, prediction of clinical outcomes, and classify disease. Virtual staining of tissue and automation of electron microscopy imaging are emerging applications with particular significance for nephropathology. The practice of medicine is changing with the development of new AI methods of machine learning. Coupled with rapid improvements in computer processing, these AI-based systems are already improving the accuracy and efficiency of diagnosis and treatment across various specializations (Table 1). The increasing focus on AI in radiology has led to some experts suggesting that someday AI may even replace radiologists. These suggestions raise the question of whether AI-based systems will eventually replace physicians in some specializations or will augment the role of physicians without actually replacing them. AI-based systems will augment physicians and are unlikely to replace the traditional physician–patient relationship.

kidney pathologies

CONCLUSION

In conclusion, the utilization of AI for non-invasive diagnosis in kidney pathologies holds immense promise in improving early detection, accuracy, and patient outcomes. With continued research and development, AI-powered diagnostic tools have the potential to revolutionize renal healthcare by providing efficient and accessible solutions for both patients and healthcare providers.

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CROSSWORD PUZZLE

C	A	S	E	S	T	U	D	I	E	S	X	T	Q	M
A	E	M	E	D	I	F	I	C	I	A	L	D	V	E
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T	0	X	E	P	E	E	R	R	E	V	I	E	W	H
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C	I	U	N	L	P	O	T	E	I	C	T	A	E	D
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POEM

BY: Oluwanifemi Odediya

'Cheers' to Revolutionary Health Care

We once leaned on the fragile might of mortals,
Tied to a realm of so-called "mending hands,"
A place where we trekked endless miles
For just a tablet of relief.
The city of "no physicians here" was crumbling swiftly.

A change was overdue—
A fresh dawn, a bold age has arrived.
No, wait—the tide had already turned.
A world that once merely pulsed a heartbeat
Now pulses with the force of innovation.

A flicker on a screen, and there's the healer,
From hulking machines to tools of care,
All shrunk down, within reach.
Technology has carved the escape route.

Once, twenty hours of surgery was horror,
But machines had mastered this horror dance,
Turning surgery into a simple feat.
From virtual realms to layers of augmentation,
The horizon of tech stretches beyond sight.

A storm of progress swept through health's domain,
Unleashing a sky full of dreams,
Enigmas, still waiting to unfold.

Tech's rise is like a confluence,
A fusion of circuits and compassionate health care,
A shattering of old limits,
A revelation,
And yes, this revolution is here to stay.

So raise a toast to what tech ignites,
The rebirth of healing's art.
We're equipped, with progress as our guard,
Delivering wholeness to the world.



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